ITA03-MP ASSIGNMENT 2

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QUESTION 1 – MOBILE COMMUNICATION OR MOBILE COMPUTING?

Total Marks: 5

1.1 Introduction

Mobile communication and mobile computing are two closely related concepts in modern information technology. Both support the use of digital devices and services beyond traditional fixed locations. However, they are not exactly the same. Mobile communication mainly focuses on transferring information between devices through communication networks, whereas mobile computing focuses on processing, storing, accessing and managing information using portable computing devices while users are moving or operating in different environments.

Mobile communication commonly uses technologies such as cellular networks, Wi-Fi, Bluetooth and other wireless communication systems. These technologies allow devices to exchange information without requiring a permanent physical cable connection. Mobile computing involves a wider range of activities, including running applications, storing information locally, accessing databases, processing data, synchronising information and continuing work when a network connection is unavailable.

It is important to understand that a device does not automatically become a mobile-computing device simply because it uses a wireless connection. Similarly, a portable computer connected to a wired network can still be used for mobile computing if it supports flexible use across different locations.

In this assignment, five devices and usage situations from a field-service utility company are examined. Each situation is classified according to its mobility, communication method and computing capabilities. An additional ambiguous example is also considered. Finally, a simple simulation is designed to compare immediate

transmission with a store-and-forward approach under different network-availability conditions.

1.2 Classification Criteria

To distinguish mobile communication from mobile computing, several criteria must be considered. A single criterion is not sufficient because mobility, connectivity and computing capabilities can differ between devices.

The following five criteria are used to classify the devices in the utility company.

1.2.1 User Mobility

User mobility refers to the ability of a person to move from one physical location to another while continuing to use a device or service.

A device used by a field technician while travelling between customer locations has a high level of user mobility. In comparison, a laptop that is moved between office buildings but is used only after being connected to a desk network has nomadic mobility rather than continuous mobility.

This criterion helps determine whether the device supports activities while the user is physically moving or whether it is simply portable and used at different stationary locations.

1.2.2 Type of Network Connection

The type of network connection identifies how a device communicates with other systems.

The main connection categories are:

- **Wired connection:** The device uses a physical cable, such as Ethernet, to connect to a network.
- **Wireless connection:** The device uses radio signals, such as cellular data or Wi-Fi, to exchange information.
- **Intermittent connection:** The device connects only when a network becomes available.
- **Offline operation:** The device can perform certain functions without an active network connection.

A wireless connection supports communication without a cable, but it does not necessarily mean that the device performs mobile computing. The device's processing and application capabilities must also be considered.

1.2.3 Disconnection Tolerance

Disconnection tolerance refers to the ability of a device or application to continue operating when the network connection is interrupted.

Some applications require a continuous network connection. If the connection is lost, the application may stop working or fail to complete its task.

Other applications are designed to continue working offline. They can store information locally and synchronise it with a central server when connectivity returns.

For example, a field technician may record meter readings while working in a location without cellular coverage. The application can save the readings locally and upload them later. This is an important feature of mobile computing.

1.2.4 Location and Context Awareness

Location awareness refers to the ability of a device or application to use information about the user's physical location or surrounding environment.

Examples include:

- GPS-based location tracking.
- Displaying nearby service requests.
- Identifying the nearest utility maintenance team.
- Recording the geographical location of a meter reading.
- Providing location-specific instructions to field technicians.

Context awareness can also include information such as time, network availability, device status and the user's current activity.

A system that uses location or context information can provide more relevant services and improve the efficiency of field operations.

1.2.5 Computing and Local Resource Capabilities

This criterion examines the device's ability to process, store and manage information locally.

A mobile-computing device generally has processing and storage capabilities that allow it to execute applications, maintain local records and perform tasks independently of a remote server for at least some period.

For example, a ruggedised tablet may contain a processor, memory, storage and applications that allow technicians to view work orders, enter information and save updates.

A communication handset may mainly provide voice or push-to-talk communication, with limited general-purpose computing functionality.

1.3 Classification of the Five Given Cases

The five devices or situations supplied in the assignment are classified below using the criteria discussed in Section 1.2.

1.3.1 Case (a): Ruggedised Tablet Fetching Work Orders Over Cellular Data

A ruggedised tablet is used by a field technician to retrieve work orders from the utility company's central system through a cellular data connection.

Classification: Mobile device – primarily mobile computing, supported by mobile communication.

Explanation:

The ruggedised tablet is designed to operate in different physical locations. A technician can carry it while travelling to customer premises, substations or other field-service locations. This demonstrates user mobility.

The device uses a cellular network to communicate with the central server. The cellular connection allows the technician to retrieve work orders without requiring a wired connection.

The tablet also has computing capabilities. It can display work orders, execute field-service applications, process information and potentially store records locally. Therefore, its main purpose is not simply to communicate but also to support computing activities.

The cellular network provides the communication facility, while the tablet application provides the computing functionality.

Reason for classification: The device combines mobility, wireless communication and application processing. It is therefore primarily a mobile-computing device that depends on mobile communication to access remote information.

1.3.2 Case (b): Laptop Connected to Ethernet at Different Offices

A laptop is moved between different offices. At each office, it is connected to the local network using an Ethernet cable.

Classification: Nomadic computing device – primarily mobile computing, using wired communication.

Explanation:

The laptop is portable and can be transported between different office locations. However, it is connected through a wired Ethernet connection when it is being used.

This means that the laptop does not rely on a wireless network for its communication in the described situation. It must be physically connected to the network at each office.

Despite using a wired connection, the laptop remains a computing device. It can run applications, process data, store files and access organisational systems.

The important distinction is between portability and continuous mobility. The laptop is portable because it can be moved between offices, but its network access depends on a physical connection.

Reason for classification: This is an example of nomadic computing. It supports computing at different locations, but it does not demonstrate wireless mobile communication in the given situation.

1.3.3 Case (c): Fixed Wireless Router at a Remote Substation

A wireless router is installed permanently at a remote electrical substation to provide network connectivity.

Classification: Fixed wireless communication – primarily wireless communication rather than mobile computing.

Explanation:

The router is installed at a fixed geographical location. It does not normally move from one location to another during operation.

Although the router uses a wireless connection, it is not a mobile device in the usual sense. Its primary function is to provide network access to equipment at the remote substation.

The router may use a cellular or other wireless link to connect the substation to the utility company's network. It forwards information between connected devices and the wider network.

The presence of a wireless connection does not automatically make the router a mobile-computing device. Its primary role is communication infrastructure rather than providing a portable environment for users to perform computing tasks.

Reason for classification: The router provides fixed wireless connectivity. It is an example of wireless communication infrastructure rather than a typical mobile-computing device.

1.3.4 Case (d): Push-to-Talk Handset

A field-service employee uses a push-to-talk handset to communicate instantly with other employees or a control centre.

Classification: Mobile communication device – primarily mobile communication.

Explanation:

A push-to-talk handset allows employees to communicate through short voice transmissions. The user can carry the handset while moving between different field locations.

The device uses a communication network or radio link to transmit voice information. Its main purpose is to provide immediate communication between team members.

Although modern handsets may contain some processing capabilities, their main function in this situation is voice communication rather than general-purpose computing.

For example, a technician can press a button to speak to the control centre and receive instructions from another employee. This supports coordination during maintenance and emergency operations.

Reason for classification: The device is portable and uses wireless communication, but its primary function is transmitting and receiving voice messages. Therefore, it is mainly a mobile-communication device.

1.3.5 Case (e): Tablet Application That Records Meter Readings Offline and Synchronises Later

A field technician uses a tablet application to record meter readings. The application can save information when there is no network connection and synchronise the stored information with the central system when connectivity becomes available.

Classification: Mobile device – primarily mobile computing, supported by intermittent mobile communication.

Explanation:

This case demonstrates an important feature of mobile computing: the ability to continue performing useful tasks even when a network connection is unavailable.

The technician can visit a customer's premises and enter the meter reading into the tablet application. If cellular coverage is unavailable, the application stores the reading in the device's local storage.

When the device reconnects to the network, the application synchronises the saved information with the central server.

This approach is commonly known as offline-first operation or store-and-forward communication.

The application must manage locally stored information, identify records that have not yet been uploaded, retry failed transmissions and confirm successful synchronisation. It may also need to detect conflicting updates.

Reason for classification: The tablet supports local data processing, storage and application execution independently of continuous network availability. Therefore, it is primarily a mobile-computing application that uses mobile communication when the network is available.

1.4 Summary of Classification

The following table summarises the classification of the five cases.

Case	Device or situation	Mobility classification	Main classification	Reason
(a)	Ruggedised tablet using cellular data	Mobile	Mobile computing	Portable application device using cellular communication
(b)	Laptop connected to Ethernet at different offices	Nomadic	Mobile computing	Portable computing device using wired network access
(c)	Fixed wireless router at a remote substation	Fixed wireless	Mobile communication / wireless communication infrastructure	Provides wireless network connectivity at a fixed location
(d)	Push-to-talk handset	Mobile	Mobile communication	Mainly supports voice transmission and reception
(e)	Tablet recording meter readings offline	Mobile	Mobile computing	Stores and processes information locally and synchronises later

The classification shows that wireless connectivity and mobile computing are related but different concepts. A device may use wireless communication without being a

mobile-computing device, and a portable computing device may use a wired connection.

1.5 Additional Ambiguous Case

1.5.1 Proposed Case: Computer System Installed in a Service Vehicle

An additional example is a computer system installed inside a utility company's maintenance vehicle. The system is connected to the company's central network through a cellular connection. It receives maintenance assignments, displays the vehicle's location and allows the technician to enter service information.

If the vehicle enters an area without network coverage, the system stores the information locally. Once cellular connectivity returns, the stored records are synchronised with the central server.

This example is considered ambiguous because the computer system is physically installed in a vehicle rather than carried directly by the user. It also combines communication, computing and location-based functionality.

1.5.2 Classification

Classification: Mobile computing supported by mobile communication.

1.5.3 Justification

The system is mobile because it operates inside a vehicle that travels between different locations. It uses cellular communication to exchange information with the central system.

It also performs computing tasks. It receives and displays assignments, processes location information, stores maintenance records and allows users to enter updates.

The system can continue operating when the network is unavailable because it stores information locally. This demonstrates disconnection tolerance.

Therefore, even though the system is installed inside a vehicle, it should be classified as a mobile-computing system rather than simply a communication device.

This example demonstrates that mobility does not always depend on whether a device is handheld. A computing system installed in a moving vehicle can also be considered mobile computing.

1.6 Simulation: Immediate-Send and Store-and-Forward Approaches

1.6.1 Objective of the Simulation

The purpose of this simulation is to examine how network availability affects the delivery of meter-reading updates from field technicians to the utility company's central server.

Two transmission approaches are compared:

1. **Immediate-send approach:** The application attempts to transmit each meter reading immediately. If the network is unavailable at that moment, the transmission fails and the record is not automatically retained for a later attempt.
2. **Store-and-forward approach:** The application stores each meter reading locally when transmission is unsuccessful. It retries transmission when network connectivity becomes available.

The simulation considers three network-availability levels:

- 90% availability.
- 60% availability.
- 30% availability.

A total of 1,000 meter-reading records are considered for each availability condition.

1.6.2 Simulation Assumptions

To make the comparison meaningful, the following assumptions are used:

1. A total of 1,000 independent meter-reading records are generated.
2. Each record contains a valid meter-reading update.
3. Network availability represents the probability that the network is available during a transmission attempt.
4. In the immediate-send approach, each record receives only one transmission attempt.
5. In the store-and-forward approach, a failed record is retained in local storage and retried.
6. The local storage capacity is sufficient to retain all 1,000 records.
7. The network eventually becomes available again, allowing stored records to be transmitted.
8. Each retry is made at the next simulated time interval.
9. Permanent server failure, device damage and local-storage failure are excluded.

10. The simulation uses independent network-availability trials. Actual network outages may be correlated and last for several consecutive time intervals.

These assumptions are important because the availability percentage alone does not describe the duration or pattern of network outages.

1.6.3 Immediate-Send Approach

In the immediate-send approach, the tablet attempts to send each meter reading as soon as it is entered.

If the network is available, the record is transmitted successfully. If the network is unavailable, the transmission fails.

The expected number of successful transmissions can be estimated using:

Expected successful transmissions = Total records × Network availability

The expected number of failed transmissions is:

Expected failed transmissions = Total records − Expected successful transmissions

For 1,000 records, the expected results are calculated as follows.

At 90% network availability:

Successful transmissions:

= 1,000 × 0.90

= 900 records

Failed transmissions:

= 1,000 − 900

= 100 records

At 60% network availability:

Successful transmissions:

= 1,000 × 0.60

= 600 records

Failed transmissions:

= 1,000 − 600

= 400 records

At 30% network availability:

Successful transmissions:

= 1,000 × 0.30

= 300 records

Failed transmissions:

= 1,000 − 300

= 700 records

The expected immediate-send results are summarised below.

Network availability	Total records	Expected delivered	Expected failed
90%	1,000	900	100
60%	1,000	600	400
30%	1,000	300	700

The results demonstrate that the immediate-send approach becomes less reliable as network availability decreases. At 30% availability, approximately 700 out of 1,000 records are expected to fail during their initial transmission attempts.

1.6.4 Store-and-Forward Approach

In the store-and-forward approach, the application does not discard a record when the network is unavailable.

Instead, it follows these steps:

1. The technician enters a meter reading.
2. The application saves the record in local storage.
3. The application checks whether a network connection is available.
4. If the network is available, the application sends the record to the central server.
5. If transmission fails, the record remains in the local queue.
6. The application retries the transmission during a later interval.
7. After successful transmission, the record is marked as synchronised.
8. The record is removed from the pending queue only after successful confirmation.

This approach reduces the possibility of losing information because of temporary network interruptions.

Under the assumption that connectivity eventually returns and the queue has sufficient capacity, all 1,000 records can eventually be delivered.

Network availability	Total records	Expected initial transmissions	Records initially queued	Eventual delivered	Permanent loss
90%	1,000	900	100	1,000	0
60%	1,000	600	400	1,000	0
30%	1,000	300	700	1,000	0

The table represents expected results under the stated assumptions. The number of records initially queued is expected to increase as network availability decreases. However, the records are retained and can be delivered later.

1.6.5 Comparison of the Two Approaches

The following table compares the main characteristics of immediate-send and store-and-forward transmission.

Comparison factor	Immediate-send	Store-and-forward
Transmission method	Sends records immediately	Stores failed records and retries
Local storage requirement	Usually low	Requires sufficient local storage
Behaviour during disconnection	Transmission fails	Records remain queued
Data-loss risk	Higher during network interruptions	Lower if local storage and retry mechanisms work correctly
Delivery time	Usually immediate when connected	May be delayed until connectivity returns
Application complexity	Relatively simple	Requires queue and synchronisation management
Suitability for field work	Less suitable for unstable coverage	More suitable for intermittent connectivity

The immediate-send approach provides low delay when a reliable connection is available. However, it is unsuitable as the only transmission method when field technicians frequently work in areas with poor network coverage.

Store-and-forward introduces additional application complexity, but it improves the reliability of data collection. This is especially important for utility companies because missing meter readings can affect billing, maintenance records and operational decisions.

1.6.6 Estimating Mean Delivery Delay

Network availability affects not only whether records are delivered but also how long delivery takes.

For the store-and-forward simulation, assume that each record is checked once per time interval and that network availability in each interval is independent.

Let:

- (p) = probability that the network is available during a transmission attempt.
- (1-p) = probability that the network is unavailable.
- (D) = number of failed attempts before a successful attempt.

The expected number of failed attempts before success is:

[
E(D)=\frac{1-p}{p}
]

If each retry interval represents one minute, the expected delay is:

Network availability	Expected failed attempts before success	Expected delivery delay
90%	0.111	0.111 minutes
60%	0.667	0.667 minutes
30%	2.333	2.333 minutes

These values are theoretical expected delays under the independent-availability model, not measured results from a real cellular network. A real network may experience longer continuous outages, so actual delivery delays can be higher.

For the immediate-send approach, records that succeed on their first attempt have approximately zero additional waiting time in this simplified model. Records that fail are

lost rather than delayed, so a mean delivery delay for failed records cannot be calculated unless a retry mechanism is introduced.

1.6.7 Simulation Implementation

The following Python program can be used to simulate the two approaches. It generates 1,000 records for each network-availability level and uses a fixed random seed so that the results can be reproduced.

```
import random
import statistics


TOTAL_RECORDS = 1000
AVAILABILITY_LEVELS = [0.90, 0.60, 0.30]
SEED = 42


def simulate_immediate_send(probability):
    random.seed(SEED)

    delivered = 0
    failed = 0

    for record in range(TOTAL_RECORDS):
        if random.random() < probability:
            delivered += 1
        else:
            failed += 1

    return delivered, failed



def simulate_store_and_forward(probability):
```

```
    random.seed(SEED + 1)

    delivery_delays = []

    for record in range(TOTAL_RECORDS):
        delay = 0

        while True:
            if random.random() < probability:
                delivery_delays.append(delay)
                break

            delay += 1

    delivered = len(delivery_delays)
    lost = TOTAL_RECORDS - delivered

    mean_delay = statistics.mean(delivery_delays)

    return delivered, lost, mean_delay


print("NETWORK SIMULATION RESULTS")
print("---------------------------------")

for probability in AVAILABILITY_LEVELS:

    immediate_delivered, immediate_failed = (
```

```
    simulate_immediate_send(probability)
)

stored_delivered, stored_lost, mean_delay = (
    simulate_store_and_forward(probability)
)

print(f"\nNetwork availability: {probability * 100:.0f}%")

print("Immediate-send:")
print("Delivered:", immediate_delivered)
print("Failed:", immediate_failed)

print("Store-and-forward:")
print("Delivered:", stored_delivered)
print("Lost:", stored_lost)
print("Mean delivery delay:", round(mean_delay, 3), "minutes")
```

Note: The program uses a simplified model in which each retry interval has an independent probability of network availability. The numerical output from the program should be recorded as the simulation result. The expected values calculated in the previous sections are analytical estimates and should not be presented as the program's exact random output.

1.6.8 Measures Used to Evaluate the Simulation

The simulation uses the following measures:

1. Number of delivered records

The total number of meter-reading records successfully received by the central server.

2. Number of failed or lost records

The number of records that are not delivered. In the immediate-send model, a failed initial attempt is counted as lost because no retry is performed. In the store-and-forward model, a record is counted as lost only if it is permanently unavailable.

3. Number of delayed records

The number of records that are not delivered during their initial transmission attempt but are delivered during a later retry.

4. Mean delivery delay

The average time taken for successfully delivered records to reach the server after they are generated.

It is calculated using:

[
\text{Mean Delay} =
\frac{\text{Sum of delivery delays}}
{\text{Number of successfully delivered records}}
]

These measures provide a clearer comparison than simply checking whether the device has a network connection.

1.7 Inquiry and Discussion

1.7.1 Which Features of Mobile Computing Are Not Solved by a Wireless Link Alone?

A wireless connection allows a device to exchange information with other systems. However, it does not automatically solve all the challenges associated with mobile computing.

Several important features require support from the device, operating system and application software.

1. Offline Data Storage

A wireless link cannot guarantee that a network connection will always be available. Mobile applications must therefore provide local storage to retain important information during disconnections.

For example, the meter-reading application must save readings on the tablet before attempting synchronisation.

2. Data Synchronisation

A wireless link can transmit information, but it does not automatically determine which records need to be uploaded or whether they have already been synchronised.

The application must maintain synchronisation status and ensure that records are not accidentally lost or duplicated.

3. Conflict Management

If the same information is modified on two different devices while they are disconnected, the application may receive conflicting updates when the devices reconnect.

The application must define a conflict-resolution policy, such as checking timestamps, comparing record versions or requesting manual confirmation.

4. Security and Authentication

Wireless transmission alone does not guarantee that information is protected. Mobile applications must use appropriate authentication, authorisation and encryption mechanisms.

This is particularly important for utility companies because work orders, customer information and meter readings may be confidential.

5. Resource Management

Mobile devices often operate using batteries and have limited storage, processing capacity and network resources compared with large servers.

Applications must manage battery usage, memory consumption, storage space and background network activity efficiently.

6. Location and Context Management

A wireless connection does not automatically provide accurate information about the user's location or activity. Applications may need GPS, location services and contextual information to support location-based operations.

Therefore, mobile computing requires application-level intelligence in addition to wireless connectivity.

1.7.2 Which Device Categories Require Application-Level Changes?

Different device categories require different levels of application support.

Ruggedised tablets: These devices may require offline caching, local work-order storage, secure access, synchronisation and conflict management. These features allow technicians to continue working when cellular connectivity is unavailable.

Ethernet-connected laptops: These devices may require local file management, network-session recovery and synchronisation if users need to continue working after moving between offices or temporarily losing connectivity. However, the described situation does not necessarily require wireless-specific changes.

Fixed wireless routers: These devices mainly require network configuration, connection monitoring, fault detection and recovery mechanisms. Application-level changes may be necessary in the equipment connected to the router, but the router itself primarily provides communication infrastructure.

Push-to-talk handsets: These devices require mechanisms for reconnecting to the communication network, managing interrupted transmissions and providing communication-status information. If they support additional applications, those applications may require further changes.

Offline meter-reading tablets: These devices require the most important application-level changes in this example. The application must support local record creation, persistent storage, queued uploads, retry handling, synchronisation confirmation and conflict management.

The comparison shows that mobile-computing features are not provided by the wireless network alone. They must be designed into the software and device architecture.

1.8 Findings and Analysis

The classification exercise demonstrates that mobile communication and mobile computing should not be treated as identical concepts.

The ruggedised tablet and offline meter-reading tablet are primarily mobile-computing devices because they support application execution and local information management while users work in different locations.

The Ethernet-connected laptop is an example of nomadic computing because it can be moved between offices but uses a wired connection in the described situation.

The fixed wireless router provides wireless communication at a stationary location. Its wireless connection does not make it a mobile-computing device.

The push-to-talk handset mainly provides mobile communication because its primary purpose is voice transmission and reception.

The simulation also demonstrates the importance of disconnection tolerance. The immediate-send approach becomes less reliable as network availability decreases. The store-and-forward approach can retain information during interruptions and deliver it later, provided that the device has sufficient storage and the connection eventually returns.

However, store-and-forward does not eliminate every problem. It can introduce delivery delays, require additional storage and create synchronisation challenges. Applications must also protect locally stored information and prevent duplicate records during repeated transmission attempts.

1.9 Conclusion

Mobile communication and mobile computing are closely related but serve different purposes. Mobile communication provides the network facilities needed to exchange information, whereas mobile computing enables users to process, store, access and manage information while operating in different locations.

The five cases demonstrate that a device must be classified according to its mobility, connection type, computing capabilities and ability to operate during network interruptions. Wireless connectivity alone is not sufficient to determine whether a device performs mobile computing.

The simulation comparison further shows that immediate transmission is vulnerable to network interruptions, while store-and-forward provides a more reliable method for collecting field information. For a utility company, an offline-capable meter-reading application is therefore a suitable approach because it allows technicians to continue working even in areas with limited network coverage.

Overall, effective mobile computing requires a combination of suitable hardware, reliable communication methods, local data management, application-level synchronisation and security mechanisms.

QUESTION 2 – SHARING THE CHANNEL: FDMA VS TDMA UNDER DIFFERENT TRAFFIC

Total Marks: 6

2.1 Introduction

Multiple Access Techniques allow several users to share a common communication medium. In mobile communication systems, the available radio spectrum is limited. Therefore, it is important to allocate the available frequency and time resources efficiently.

Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA) are two important techniques used to share communication channels among multiple users.

FDMA divides the available frequency spectrum into separate frequency channels. Each user is assigned a particular frequency channel for communication. Different users can communicate simultaneously because they use different frequency bands.

TDMA divides communication into repeating time slots. Users share the same carrier frequency, but each user is allocated a specific time slot during which transmission can take place. This allows multiple users to use the same frequency channel at different times.

The main difference is that FDMA separates users by frequency, whereas TDMA separates users by time.

In this assignment, a cellular system with 2 MHz of available spectrum in each direction is considered. The capacity of FDMA and TDMA is calculated, and their performance is analysed under different traffic conditions. The analysis also considers channel overhead, traffic utilisation, blocking and wasted capacity.

2.2 Working Principles of FDMA

Frequency Division Multiple Access (FDMA) is a channel-allocation technique in which the available frequency spectrum is divided into smaller frequency bands.

Each user is allocated a dedicated frequency channel for the duration of the communication session. Other users communicate using different frequency channels.

For example, if a cellular system has 2 MHz of available spectrum and each FDMA channel requires 25 kHz, the maximum nominal number of channels can be calculated by dividing the total bandwidth by the channel bandwidth.

Formula:

Number of FDMA channels = Total available bandwidth / Bandwidth per channel

Given:

Total bandwidth = 2 MHz = 2000 kHz

Bandwidth per channel = 25 kHz

Number of FDMA channels = 2000 / 25

Number of FDMA channels = 80

Therefore, the system can theoretically provide 80 nominal FDMA channels in each direction.

In practical systems, guard bands may be required between adjacent frequency channels to reduce interference.

2.2.1 Advantages of FDMA

1. Simple implementation: Each user receives a dedicated frequency channel, making resource allocation straightforward.
2. Continuous transmission: Users can transmit continuously while their frequency channel is assigned.
3. Low scheduling complexity: The system does not need to coordinate every transmission through a time-slot schedule.

4. Suitable for continuous traffic: FDMA can be useful for applications such as traditional voice communication, where a continuous channel is required.
5. Predictable resource allocation: A user receives a specific frequency resource during the communication session.

2.2.2 Disadvantages of FDMA

1. Frequency wastage: An assigned channel may remain unused during periods when the user has no information to transmit.
2. Guard-band overhead: Frequency separation may be required to reduce interference between adjacent channels.
3. Limited flexibility: A frequency channel allocated to one user may not be easily shared with another user during idle periods.
4. Blocking: New users may be rejected when all frequency channels are occupied.
5. Limited suitability for bursty traffic: Applications that transmit small amounts of data occasionally may waste a dedicated channel.

2.3 Working Principles of TDMA

Time Division Multiple Access (TDMA) divides a carrier frequency into multiple time slots. Each user receives a specific time slot during which they can transmit or receive information.

A TDMA system repeatedly cycles through a sequence of time slots called a frame. Users transmit during their allocated slots, and the system synchronises transmissions to prevent overlapping transmissions.

In the assignment scenario, each TDMA carrier has a bandwidth of 200 kHz and is divided into eight time slots.

Therefore, one carrier can support eight nominal user slots.

The advantage of TDMA is that several users share the same frequency carrier. The available transmission time is divided among them.

2.3.1 Advantages of TDMA

1. Shared frequency resources: Multiple users can use one carrier at different times.
2. Efficient digital transmission: TDMA can support digital communication systems.
3. Flexible scheduling: Time slots can be assigned according to traffic requirements in systems that support dynamic allocation.

4. Potentially better burst handling: Short transmissions can use allocated slots without requiring continuous frequency occupation, provided that the scheduling system supports such sharing.
5. Multiple users per carrier: Several users can share one frequency carrier through separate time slots.

2.3.2 Disadvantages of TDMA

1. Time synchronisation: The system must coordinate transmissions accurately.
2. Guard-time overhead: Small time gaps may be required between slots to prevent overlap.
3. Control overhead: Synchronisation and control information consume part of the available transmission capacity.
4. Scheduling requirements: The system must manage time-slot assignments.
5. Delay: A user may need to wait until the next allocated time slot before transmitting.

2.4 Given System Parameters

The assignment provides the following system parameters:

Parameter	Given Value
Total bandwidth per direction	2 MHz
FDMA channel bandwidth	25 kHz
TDMA carrier bandwidth	200 kHz
Number of TDMA slots per carrier	8
Nominal FDMA channels	To be calculated
Nominal TDMA user slots	To be calculated

The calculations assume that the full 2 MHz is available for the relevant direction of communication. The uplink and downlink are treated separately because the assignment specifies 2 MHz per direction.

2.5 Nominal Capacity Calculation

2.5.1 FDMA Capacity

The total available bandwidth is:

Total bandwidth = 2 MHz = 2000 kHz

Each FDMA channel requires:

Channel bandwidth = 25 kHz

The nominal number of channels is calculated as follows:

Number of FDMA channels = Total bandwidth / Channel bandwidth

Number of FDMA channels = 2000 / 25

Number of FDMA channels = 80

Therefore, the system can theoretically provide 80 nominal FDMA channels in each direction.

2.5.2 TDMA Capacity

The total available bandwidth is:

Total bandwidth = 2000 kHz

Each TDMA carrier requires:

Carrier bandwidth = 200 kHz

The number of available carriers is:

Number of carriers = 2000 / 200

Number of carriers = 10

Each carrier contains eight time slots.

Therefore:

Number of TDMA user slots = Number of carriers × Slots per carrier

= 10 × 8

Number of TDMA user slots = 80

Therefore, the TDMA system also provides 80 nominal user slots in each direction.

Both techniques provide the same nominal capacity of 80 user channels or slots per direction. However, actual usable capacity may be lower because practical systems require resources for guard bands, guard times, synchronisation and control information.

2.6 Estimating Usable Capacity After Overhead

The nominal capacity calculation does not consider the resources required for system operation. In practical communication systems, some bandwidth or transmission time must be reserved to prevent interference and maintain synchronisation.

The assignment requires an estimate of usable capacity after considering guard bands, guard times and control overhead.

The following assumptions are used for an illustrative calculation. These overhead values are not specified in the assignment, so they are treated as design assumptions rather than fixed standards parameters.

2.6.1 FDMA Guard-Band Assumption

For FDMA, assume that 1 kHz of each 25 kHz channel is reserved as an effective guard-band allowance.

The useful bandwidth per channel becomes:

Useful bandwidth = 25 − 1

Useful bandwidth = 24 kHz

The total useful bandwidth is:

Total useful bandwidth = 80 × 24

Total useful bandwidth = 1920 kHz

The percentage of useful bandwidth is:

Efficiency = (1920 / 2000) × 100

FDMA efficiency = 96%

The effective guard-band allowance is:

Guard-band allowance = 2000 − 1920

Guard-band allowance = 80 kHz

Thus, under this assumption, 96% of the total spectrum is available for useful channel bandwidth.

2.6.2 TDMA Guard-Time and Control-Overhead Assumption

For TDMA, assume that 10% of the available time-slot capacity is reserved for guard times, synchronisation and control information.

The usable proportion is:

Usable proportion = 100% − 10%

Usable proportion = 90%

The nominal TDMA capacity is 80 slots.

The equivalent usable slot capacity is:

Usable capacity = 80 × 0.90

Usable capacity = 72 full-slot equivalents

Therefore, under this illustrative assumption, TDMA provides 72 full-slot equivalents of useful transmission capacity.

The reserved capacity is:

Reserved capacity = 80 − 72

Reserved capacity = 8 slot equivalents

2.6.3 Capacity Comparison

Parameter	FDMA	TDMA
Total bandwidth	2 MHz	2 MHz
Basic resource size	25 kHz	200 kHz carrier
Number of carriers/channels	80 channels	10 carriers
Slots per carrier	Not applicable	8
Nominal user capacity	80	80
Assumed overhead	1 kHz per channel	10% of slot capacity
Estimated useful capacity	80 channels with 24 kHz useful bandwidth each	72 full-slot equivalents
Estimated efficiency	96%	90%

The comparison indicates that both techniques provide the same nominal user capacity under the given parameters. However, their usable capacities differ when the assumed overhead is included.

It is important to understand that the 96% FDMA and 90% TDMA figures are illustrative results based on the assumptions made in this report. Actual efficiency depends on the radio-system design, modulation, coding, synchronisation requirements and control-channel structure.

2.7 Predicted Utilisation Under Different Traffic Conditions

The performance of a multiple-access technique depends heavily on the type of traffic being transmitted.

Three main traffic conditions are considered:

1. Continuous voice traffic.
2. Bursty data traffic with approximately 10% duty cycle.
3. A mixed traffic environment containing 70% voice and 30% data.

2.7.1 Scenario 1: Continuous Voice Traffic

In continuous voice communication, a user generally needs a channel for the duration of the call.

For example, a user may maintain a voice connection for several minutes. Even if there are short pauses in speech, the connection may remain allocated.

FDMA Prediction:

FDMA is suitable for continuous voice traffic because each user receives a dedicated frequency channel. The user can transmit continuously without waiting for a specific time slot.

However, the assigned frequency channel may still be unused during silence periods or pauses in conversation.

TDMA Prediction:

TDMA can also support continuous voice communication by assigning recurring time slots to users. Each user receives transmission opportunities according to the frame structure.

The system must maintain accurate synchronisation and manage the allocated slots.

Expected Comparison:

Both FDMA and TDMA can support continuous voice traffic. FDMA offers straightforward dedicated-channel operation, while TDMA shares each carrier among several users through time slots.

Under the simplified dedicated-allocation model, both have similar nominal user capacity. Their practical performance depends on the overhead and traffic-management design.

2.7.2 Scenario 2: Bursty Data Traffic with Approximately 10% Duty Cycle

Bursty data traffic consists of short periods of transmission separated by relatively long periods of inactivity.

For example, a sensor may transmit a small data packet every few seconds or minutes.

A duty cycle of 10% means that a device is actively transmitting for approximately 10% of the observation period. The remaining 90% of the time is idle.

FDMA Prediction:

If a dedicated frequency channel remains allocated to the device during its inactive periods, a large amount of capacity may be wasted.

For example, if a device transmits for only 10% of the time, its dedicated channel remains unused for approximately 90% of the observation period.

TDMA Prediction:

TDMA can provide better flexibility when the system supports dynamic allocation of time slots. Short data transmissions can be scheduled into available slots rather than reserving a continuous frequency resource for every inactive device.

However, if each user is assigned a fixed dedicated slot throughout the entire observation period, TDMA can also waste capacity.

Therefore, the benefit depends on whether the system uses fixed or dynamic slot allocation.

Expected Comparison:

Dynamic TDMA can be more efficient for bursty data because time slots can be reassigned according to demand. FDMA may waste more capacity when channels remain dedicated to users who are inactive for long periods.

2.7.3 Scenario 3: Mixed Traffic with 70% Voice and 30% Data

In this scenario, 70% of the traffic consists of voice communication and 30% consists of data communication.

Voice traffic usually requires predictable transmission opportunities, whereas data traffic may be irregular and bursty.

FDMA Prediction:

FDMA can provide dedicated frequency channels for voice users. This offers straightforward resource allocation for continuous communication.

However, data users may not continuously use their assigned channels. If the channels cannot be reassigned dynamically, idle capacity may result.

TDMA Prediction:

TDMA can support a mixture of traffic by assigning regular time slots to voice users and scheduling data transmissions in available slots.

A flexible scheduler can allocate time resources according to current traffic demand.

Expected Comparison:

TDMA may provide greater flexibility in a mixed traffic environment when dynamic allocation is available. FDMA may remain suitable for users who require continuous dedicated channels.

A hybrid allocation system can also be considered, where voice and data traffic receive different resource-management policies.

2.8 Simulation Design: Three Contrasting Traffic Datasets

To compare FDMA and TDMA under different traffic conditions, three datasets are designed.

The purpose of these datasets is to observe how traffic behaviour affects resource utilisation, blocking and wasted capacity.

The simulation uses 80 nominal resource units for each technique. For a simplified comparison, one FDMA channel and one TDMA time slot are treated as one user resource unit.

The usable capacity assumptions from Section 2.6 can also be considered in a separate capacity-limit experiment. The first comparison should use nominal capacity so that the effects of traffic patterns can be examined independently of the overhead assumptions.

2.8.1 Dataset 1: Continuous Voice Traffic

The first dataset represents users who require continuous communication.

Dataset Characteristics:

- 80 users are available.
- Each user requests one dedicated channel.
- Each call remains active throughout the observation period.
- Calls are assumed to have long holding times.
- Users do not release their channels during the observation period.

Expected Result:

Both FDMA and TDMA can allocate resources to 80 users when all 80 nominal resources are available.

Once all resources are occupied, additional users cannot be accepted.

This scenario is expected to show high utilisation when all 80 users are active.

2.8.2 Dataset 2: Bursty Data Traffic

The second dataset represents applications that transmit short packets occasionally.

Dataset Characteristics:

- 80 devices are considered.
- Each device has an active transmission probability of approximately 10% during a given time interval.
- Each active device generates a short data packet.
- Devices remain idle during most observation intervals.
- A device may become active again during a later interval.

The expected number of active devices in one observation interval is:

Number of active devices = Total devices × Activity probability

= 80 × 0.10

Expected active devices = 8

Therefore, approximately eight devices are expected to be active during an average observation interval.

Expected Result:

If resources remain permanently dedicated to all 80 devices, the utilisation may be very low.

If dynamic time-slot allocation is used, TDMA can allocate resources to active devices and leave fewer resources reserved for inactive devices.

This scenario is useful for examining the difference between fixed resource allocation and demand-based scheduling.

2.8.3 Dataset 3: Synchronized Arrivals – Edge Case

The third dataset is designed as an edge case in which many devices attempt to transmit at the same time.

For example, a group of utility sensors may be configured to send reports at the beginning of every minute.

Dataset Characteristics:

- 80 devices are considered.
- All devices attempt to transmit at the same time.
- Each device generates one data packet.
- Each packet requires one resource unit.

- The system has 80 nominal resource units.
- A second experiment can increase the number of simultaneous devices to 100.

Expected Result:

When 80 devices transmit simultaneously and 80 resources are available, all devices can be allocated a resource under the simplified model.

When 100 devices transmit simultaneously, only 80 can be served immediately. The remaining 20 requests are blocked or delayed, depending on the queueing policy.

This scenario demonstrates that even a system with good average utilisation can experience congestion when many devices generate traffic simultaneously.

2.9 Simulation Measures

The following measures are used to compare the performance of FDMA and TDMA.

2.9.1 Resource Utilisation

Resource utilisation represents the proportion of available resources that are actively used.

Formula:

Resource Utilisation = (Busy resource units / Total resource units) × 100

For example, if 40 out of 80 resources are occupied:

Resource Utilisation = (40 / 80) × 100

Resource Utilisation = 50%

2.9.2 Blocking Rate

Blocking occurs when a new user cannot obtain a resource because all available resources are occupied.

Formula:

Blocking Rate = (Blocked requests / Total requests) × 100

For example, if 20 out of 100 requests are blocked:

Blocking Rate = (20 / 100) × 100

Blocking Rate = 20%

2.9.3 Wasted Capacity

Wasted capacity refers to resources that are reserved or allocated but remain unused.

For a fixed allocation system, it can be estimated as:

Wasted Capacity = Allocated resources − Busy resources

The wasted-capacity percentage is:

Wasted Capacity Percentage = (Wasted resources / Total resources) × 100

These measures help determine whether a multiple-access technique uses available resources efficiently.

2.10 Simulation Procedure

The simulation can be performed using the following procedure:

1. Define the total number of resource units as 80.
2. Create the three traffic datasets.
3. Generate transmission requests according to each dataset.
4. Allocate a resource to each request when a resource is available.
5. Count the number of resources currently occupied.
6. Record the number of blocked requests.
7. Calculate resource utilisation.
8. Calculate wasted capacity.
9. Repeat the experiment using FDMA and TDMA.
10. Compare the results under the same traffic conditions.

For a fair comparison, both techniques should use the same traffic arrivals, observation periods and nominal resource limits.

An important distinction must be maintained: if the experiment models fixed dedicated allocation, both techniques should retain their assigned resources during idle periods. If dynamic TDMA is tested, it should be reported as a separate scheduling experiment rather than treating it as an automatic property of all TDMA systems.

2.11 Expected Simulation Results and Result Table

The following table gives the expected qualitative behaviour. Actual numerical simulation results must be generated using the implementation and recorded separately.

Traffic Dataset	FDMA Expected Behaviour	TDMA Expected Behaviour

Continuous voice	Good support for continuous dedicated communication	Good support through recurring time slots
Bursty data	Potentially high idle-channel wastage under dedicated allocation	Potentially better efficiency with dynamic slot allocation
Synchronized arrivals	Blocking occurs if simultaneous requests exceed available channels	Blocking or delay occurs if simultaneous requests exceed available slots

The following table should be completed using the actual simulation output.

Dataset	**Technique**	**Resource Utilisation (%)**	**Blocked Requests**	**Wasted Resource Units**
Continuous voice	FDMA	Enter measured result	Enter measured result	Enter measured result
Continuous voice	TDMA	Enter measured result	Enter measured result	Enter measured result
Bursty data	FDMA	Enter measured result	Enter measured result	Enter measured result
Bursty data	TDMA	Enter measured result	Enter measured result	Enter measured result
Synchronized arrivals	FDMA	Enter measured result	Enter measured result	Enter measured result
Synchronized arrivals	TDMA	Enter measured result	Enter measured result	Enter measured result

The numerical values should not be guessed because the assignment expects evidence from the student's own experiment.

2.12 Analysis of Simulation Results

The simulation results should be analysed by considering the relationship between traffic behaviour and resource allocation.

For continuous voice traffic, both techniques can provide predictable access because users require resources for relatively long periods. In this situation, the difference in performance may be influenced more by the overhead and implementation details than by the basic allocation principle.

For bursty data traffic, fixed dedicated allocation can result in significant wasted capacity. A resource remains assigned to a device even when the device has no information to transmit. A TDMA system that supports dynamic scheduling can make better use of available time slots by assigning them to active users.

For synchronized arrivals, both techniques can experience blocking when the number of simultaneous requests exceeds the available capacity. This shows that average traffic levels alone are not sufficient to evaluate a communication system. The timing of traffic arrivals must also be considered.

The results should therefore be interpreted using utilisation, blocking and wasted capacity together. A technique that achieves high utilisation may still experience unacceptable blocking, while a technique with low blocking may reserve too many resources.

2.13 Inquiry: Why Do FDMA and TDMA Behave Differently?

The performance of FDMA and TDMA differs because the techniques allocate communication resources in different ways.

FDMA allocates frequency resources to users. Once a frequency channel is assigned, it may remain reserved for that user throughout the communication session. This is suitable for continuous communication but may be inefficient when users transmit data only occasionally.

TDMA allocates transmission opportunities in time. It allows multiple users to share the same carrier frequency through separate time slots.

When dynamic scheduling is available, TDMA can assign time slots according to current traffic demand. This can improve resource utilisation when users generate short and irregular transmissions.

However, TDMA also requires synchronisation, guard times and control information. These requirements consume part of the available capacity.

The difference between the techniques is therefore influenced by:

1. The duration of user activity.
2. The frequency of transmission requests.
3. Whether resources are fixed or dynamically allocated.
4. The amount of guard-band or guard-time overhead.
5. The scheduling and synchronisation requirements.
6. The number of simultaneous users.

7. The distribution of traffic arrivals over time.

The main conclusion is that no single multiple-access technique is automatically the most efficient under every traffic condition. The best choice depends on the characteristics of the application and the resource-allocation method.

2.14 Design Decision: Campus Telemetry System

The assignment considers a campus telemetry system in which 5,000 devices send small reports.

Approximately 90% of the devices are expected to transmit a 200-byte report every minute.

The question is whether dedicated FDMA or TDMA channel allocation is justified for this type of traffic.

2.14.1 Number of Active Devices

Total devices = 5000

Active proportion = 90% = 0.90

Expected active devices = 5000 × 0.90

Expected active devices = 4500 devices

Therefore, approximately 4,500 devices are expected to send reports during each one-minute reporting period.

2.14.2 Data Generated Per Minute

Each device generates 200 bytes.

Total data generated = 4500 × 200

Total data generated = 900,000 bytes per minute

This is equivalent to approximately 900 kB per minute using decimal units.

2.14.3 Data Rate Calculation

One byte contains eight bits.

Total data in bits = 900,000 × 8

Total data = 7,200,000 bits

The reporting interval is 60 seconds.

Average payload data rate = 7,200,000 / 60

Average payload data rate = 120,000 bits per second

Therefore:

Average payload data rate = 120 kbps

This represents the average payload data rate before protocol headers, retransmissions, acknowledgements and other overheads are included.

2.14.4 Is Dedicated FDMA or TDMA Justified?

Dedicated FDMA allocation is not an ideal default choice for this telemetry system because each device sends only a small report once per minute.

If a dedicated frequency channel is reserved for every device, most of the channel capacity may remain unused between reports.

Dedicated TDMA allocation can have a similar problem if each device is assigned a permanent time slot even though it transmits only occasionally.

A dynamically scheduled TDMA approach may be more suitable because time slots can be allocated when devices have reports to send. This can reduce the amount of time reserved for inactive devices.

A shared random-access or scheduled-access approach may also be considered. Devices can request transmission opportunities or transmit according to a coordinated reporting schedule.

However, the system must consider the possibility that many devices may transmit simultaneously. If all 4,500 active devices attempt to send their reports at exactly the same time, the system may experience congestion.

To reduce this problem, the reporting schedule can be distributed across the one-minute interval. For example, devices can be assigned different transmission times or groups can be scheduled in separate intervals.

2.14.5 Recommended Design

For the campus telemetry system, I would recommend a shared or dynamically scheduled communication approach rather than permanently dedicating an FDMA channel or TDMA slot to every device.

The main reasons are:

1. Each device sends a relatively small amount of information.
2. The devices are inactive for most of the reporting interval.
3. Dedicated resources may remain unused between transmissions.
4. Dynamic scheduling can allocate resources according to current demand.

5. Reporting times can be distributed to avoid synchronized transmission peaks.
6. The system can allocate additional resources temporarily when traffic increases.

The design should also include packet acknowledgements, retransmission handling and queue management to improve reliability.

A dedicated allocation may still be appropriate for devices with strict latency, guaranteed service or continuous communication requirements. However, for small periodic telemetry reports, permanent dedicated allocation is likely to waste resources.

2.15 Limitations and Assumptions

The calculations and predictions in this section are based on the parameters provided in the assignment and additional assumptions stated for the overhead comparison.

The following limitations should be considered:

1. The exact FDMA guard-band requirement is not provided. The 1 kHz allowance is an illustrative assumption.
2. The exact TDMA guard-time and control overhead are not provided. The 10% allowance is an illustrative assumption.
3. The nominal capacity calculation does not include all practical radio-system constraints.
4. The traffic scenarios simplify real-world communication behaviour.
5. Actual blocking depends on arrival rates, call durations, queueing policies and resource-allocation rules.
6. Dynamic TDMA performance depends on the scheduler and is not guaranteed merely by using TDMA.
7. The telemetry data-rate calculation excludes communication-protocol overhead and retransmissions.
8. Actual performance should be validated through simulation and experimental results.

These limitations are important because theoretical capacity does not always represent the number of users that a real communication system can reliably support.

2.16 Conclusion

FDMA and TDMA are two important multiple-access techniques used to share limited radio spectrum among multiple users.

For the given system, FDMA provides 80 nominal frequency channels, while TDMA provides 10 carriers with eight slots each, giving 80 nominal user slots. After applying the illustrative overhead assumptions, FDMA provides 96% useful bandwidth and TDMA provides 90% usable slot capacity.

The traffic analysis shows that both techniques can support continuous voice communication. However, bursty data traffic may result in wasted capacity when resources are permanently dedicated to inactive users. A dynamically scheduled TDMA system can improve efficiency by allocating time slots according to current demand.

The synchronized-arrival scenario demonstrates that both techniques can experience blocking when the number of simultaneous transmission requests exceeds available capacity.

For the campus telemetry system, approximately 4,500 devices are expected to send 200-byte reports per minute, producing an average payload data rate of approximately 120 kbps. Permanently dedicating a channel or slot to every device is not recommended because the devices transmit small amounts of information and remain inactive for most of the time.

A shared or dynamically scheduled resource-allocation approach is therefore more appropriate, provided that it can satisfy the system's delay, reliability and capacity requirements

QUESTION 3 – CDMA: ORTHOGONALITY, SPREADING AND THE NEAR-FAR PROBLEM

Total Marks: 7

3.1 Introduction

Code Division Multiple Access (CDMA) is a multiple-access technique used in mobile communication systems in which multiple users share the same frequency band at the same time. Unlike Frequency Division Multiple Access (FDMA), which separates users by frequency, and Time Division Multiple Access (TDMA), which separates users by time, CDMA separates users using unique spreading codes.

In CDMA, each user is assigned a spreading code that is used to spread the original information signal over a wider bandwidth. The receiver uses the same spreading code to recover the original information from the received signal.

CDMA allows multiple users to transmit simultaneously using the same frequency spectrum. However, the system must manage interference between users. The performance depends on code properties, synchronisation, transmission power and the number of active users.

One important challenge in CDMA is the near-far problem. This occurs when a receiver receives signals from users at significantly different power levels. A nearby user may

transmit a much stronger signal than a distant user. The stronger signal can interfere with the weaker signal and make correct detection difficult.

In this assignment, a Direct Sequence Code Division Multiple Access (DS-CDMA) simulation is considered for a crowded stadium environment. The simulation examines spreading, despreading, orthogonal and non-orthogonal codes, processing gain and the near-far problem. Different spreading-code lengths are also compared to understand their effect on system performance.

3.2 Working Principle of CDMA

CDMA allows multiple users to share the same frequency band simultaneously by assigning a unique code to each user.

Each user multiplies their original information signal by a spreading code. This process increases the bandwidth occupied by the signal and makes it distinguishable from signals transmitted by other users.

At the receiver, the received signals are combined. The receiver applies the spreading code belonging to the intended user to recover the original information.

The basic CDMA communication process consists of the following stages:

1. Generation of the original information bits.
2. Conversion of information bits into digital symbols.
3. Multiplication of symbols by the spreading code.
4. Transmission of the spread signal through the communication channel.
5. Combination of signals from multiple users.
6. Despreading at the receiver.
7. Decision-making to recover the transmitted information bits.

3.3 Direct Sequence Spread Spectrum (DS-SS)

Direct Sequence Spread Spectrum is a technique in which the original information signal is multiplied by a high-rate spreading sequence.

The spreading sequence contains multiple chips for each information bit. A chip is a smaller element of the spreading code used to spread the original information signal.

For example, if a spreading code contains eight chips, one information bit is represented using eight spreading chips.

The original information signal therefore occupies a wider bandwidth than the unspread information signal.

The spreading process provides several benefits:

- It allows multiple users to share the same frequency band.
- It helps distinguish users through their unique spreading codes.
- It provides processing gain.
- It can improve resistance to certain forms of interference.
- It supports simultaneous communication by multiple users.

3.4 Spreading and Despreading Process

3.4.1 Spreading

Spreading is the process of multiplying the original information signal by a spreading code.

In a simplified binary phase-shift keying (BPSK) system, information bits are represented using two values:

- Bit 1 is represented by +1.
- Bit 0 is represented by −1.

The spreading code also contains values of +1 and −1.

The transmitted spread signal is calculated by multiplying the information symbol by each chip of the spreading code.

Formula:

Transmitted spread signal = Information symbol × Spreading code

For example, consider an information bit represented by +1 and the following spreading code:

Spreading code = [+1, −1, +1, +1, −1, +1, −1, −1]

The spread signal becomes:

Spread signal = [+1, −1, +1, +1, −1, +1, −1, −1]

If the information symbol is −1, the resulting spread signal becomes:

Spread signal = [−1, +1, −1, −1, +1, −1, +1, +1]

Therefore, the same spreading code can represent different information symbols.

3.4.2 Despreading

Despreading is the reverse process performed at the receiver.

The receiver multiplies the received signal by the spreading code assigned to the intended user.

If the correct spreading code is used and the received signal is sufficiently free from interference, the original information can be recovered.

The receiver then calculates a decision value. If the decision value is positive, the receiver detects bit 1. If the decision value is negative, it detects bit 0.

The receiver can therefore recover the intended user's information while reducing the contribution of other users whose signals use different spreading codes.

3.4.3 Decision Process

After despreading, the receiver applies a decision rule.

Decision rule:

- If the despread output is greater than zero, detect bit 1.
- If the despread output is less than zero, detect bit 0.

The detected bit is compared with the original transmitted bit to determine whether an error has occurred.

3.5 Processing Gain

Processing gain is an important characteristic of spread-spectrum communication systems.

It represents the improvement obtained by spreading the signal over a wider bandwidth and then recovering it using the correct spreading code.

The processing gain can be estimated using the ratio between the spread bandwidth and the original information bandwidth.

For a simple DS-CDMA system, the processing gain is related to the spreading-code length.

Formula:

Processing Gain = Spreading Code Length

The processing gain in decibels is calculated as:

\[G_p=10\log_{10}(N) \]

Where:

- \(G_p\) = processing gain in decibels.
- \(N\) = number of chips per information bit.

For example, if the spreading-code length is 8:

$$ G_p=10\log_{10}(8) $$

$$ \boxed{G_p=9.03\text{ dB}} $$

A larger spreading-code length provides a higher processing gain.

3.6 Comparison of Different Spreading-Code Lengths

The assignment requires the use of different spreading-code lengths, such as 8, 16 and 64.

The processing gain for each code length is calculated below.

For code length 8:

$$ G_p=10\log_{10}(8) $$

$$ \boxed{G_p=9.03\text{ dB}} $$

For code length 16:

$$ G_p=10\log_{10}(16) $$

$$ \boxed{G_p=12.04\text{ dB}} $$

For code length 64:

$$ G_p=10\log_{10}(64) $$

$$ \boxed{G_p=18.06\text{ dB}} $$

The results are summarised below.

Spreading-code length	Processing gain (linear)	Processing gain (dB)
8	8	9.03 dB
16	16	12.04 dB
64	64	18.06 dB

The comparison shows that increasing the spreading-code length increases the processing gain.

A higher processing gain can help reduce the effect of interference after despreading. However, a longer code also requires more chip transmissions for each information bit. Therefore, there is a trade-off between spreading-code length, transmission resources and system performance.

3.7 Orthogonal and Non-Orthogonal Spreading Codes

The performance of a CDMA system depends significantly on the correlation properties of its spreading codes.

Two important categories are orthogonal spreading codes and non-orthogonal spreading codes.

3.7.1 Orthogonal Spreading Codes

Orthogonal spreading codes are designed so that the correlation between different codes is zero when they are correctly aligned.

This means that when a receiver uses one user's spreading code, the contribution from another user using a different orthogonal code can ideally be removed during despreading.

Walsh codes are a common example of orthogonal spreading codes.

For example, two simple orthogonal codes are:

Code A = [+1, +1]

Code B = [+1, −1]

Their correlation is:

$$ R_{AB}=(1\times1)+(1\times-1) $$

$$ R_{AB}=1-1 $$

$$ \boxed{R_{AB}=0} $$

Since the correlation is zero, the codes are orthogonal.

In an ideal synchronous CDMA system, orthogonal codes can reduce interference between users.

However, this ideal property depends on accurate synchronisation and the absence of channel effects that destroy orthogonality.

3.7.2 Non-Orthogonal Spreading Codes

Non-orthogonal codes do not have zero correlation for every pair of users.

When multiple users transmit simultaneously, their signals may interfere with each other during despreading.

Random binary spreading codes are an example of code sequences that are generally non-orthogonal.

The correlation between two random codes may be positive, negative or zero. The amount of interference depends on their actual correlation.

As the number of simultaneous users increases, the total multiple-access interference may also increase.

This can result in a higher bit error rate and reduced communication reliability.

3.7.3 Difference Between Orthogonal and Non-Orthogonal Codes

Comparison factor	Orthogonal codes	Non-orthogonal codes
Cross-correlation	Ideally zero when aligned	Generally non-zero
User interference	Low under ideal synchronisation	More likely
Synchronisation requirement	High	Less dependent on perfect orthogonality
Multiple-user performance	Good under ideal conditions	Depends on code correlation and user loading
Example	Walsh codes	Random binary codes

3.8 DS-CDMA Simulation for a Crowded Stadium

3.8.1 Simulation Objective

The purpose of the simulation is to examine the performance of a DS-CDMA communication system in a crowded stadium where multiple users transmit information simultaneously.

In a stadium environment, a large number of users may communicate through the same radio system. If the system does not manage interference properly, the signals transmitted by different users may affect one another.

The simulation investigates three conditions:

1. Synchronous orthogonal Walsh codes.
2. Random non-orthogonal spreading codes of the same length.
3. Orthogonal codes with one user transmitting at ten times the power of another user, under imperfect synchronisation.

The simulation also examines code lengths of 8, 16 and 64 to compare processing gain and bit error rate.

3.8.2 Simulation Parameters

The following parameters are used as the basis for the simulation.

Parameter	Selected value
Communication technique	DS-CDMA
Modulation	BPSK
Spreading method	Direct Sequence Spread Spectrum
Code lengths	8, 16 and 64
Code types	Walsh and random binary
Number of users	Varied during simulation
Transmission power	Equal power in normal conditions
Near-far power ratio	10:1
Near-far power difference	10 dB
Receiver	Correlation-based matched filter
Main performance measure	Bit Error Rate
Additional measure	Code cross-correlation

The simulation should use a fixed random seed so that the results can be reproduced.

3.8.3 Simulation Condition 1: Synchronous Orthogonal Walsh Codes

In the first condition, all users transmit simultaneously using orthogonal Walsh spreading codes.

All users are assumed to be synchronised so that their spreading-code chips arrive at the receiver at the same time.

Under ideal conditions, the cross-correlation between different Walsh codes is zero.

Therefore, the receiver can recover the intended user's information while suppressing the contributions from other users.

Prediction:

The bit error rate is expected to remain relatively low as the number of users increases, provided that the number of users does not exceed the available orthogonal codes and the system maintains synchronisation.

The code length determines how many orthogonal codes can be generated. A code length of 8 supports up to 8 mutually orthogonal code sequences, a length of 16 supports up to 16, and a length of 64 supports up to 64.

The system may experience performance degradation if noise, multipath propagation or synchronisation errors affect the received signals.

3.8.4 Simulation Condition 2: Random Non-Orthogonal Codes

In the second condition, users are assigned random binary spreading codes with the same code lengths.

Unlike Walsh codes, random binary codes are not guaranteed to be orthogonal.

When users transmit simultaneously, the receiver may detect interference from other users because their spreading codes can have non-zero cross-correlation.

Prediction:

The bit error rate is expected to increase as the number of simultaneous users increases.

A larger spreading-code length may reduce the average correlation between random code sequences and provide a higher processing gain. However, it does not guarantee zero interference.

Therefore, random non-orthogonal codes are expected to perform less effectively than perfectly synchronised Walsh codes under the same idealised conditions.

3.8.5 Simulation Condition 3: Orthogonal Codes with a 10:1 Power Difference

In the third condition, users are assigned orthogonal spreading codes, but one user transmits with ten times the power of another user.

A power ratio of 10:1 corresponds to a difference of:

\[10\log_{10}(10)=10\text{ dB} \]

Therefore, the stronger user's signal is 10 dB more powerful than the weaker user's signal.

This condition is used to investigate the near-far problem.

In an ideal synchronous system with perfectly orthogonal codes and a linear channel, a power difference alone does not destroy orthogonality. The receiver can still separate the users mathematically.

However, practical wireless systems may experience timing misalignment, multipath propagation and other channel effects. These effects can cause the received spreading codes to lose their orthogonality.

When a strong user's signal has non-zero correlation with a weaker user's code, the strong signal may create significant interference during despreading.

Prediction:

Under imperfect synchronisation, the bit error rate of the weaker user is expected to increase because the strong user's signal can interfere with its detection.

This effect demonstrates why CDMA systems require power control and accurate synchronisation.

3.9 Code Correlation and Its Importance

Code correlation measures the similarity between two spreading sequences.

It is an important factor in determining how effectively a CDMA receiver can separate different users.

For two spreading codes \(C_i\) and \(C_j\), the normalised correlation can be expressed as:

\[R_{ij}=\frac{1}{N}\sum_{k=1}^{N}C_i(k)C_j(k) \]

Where:

- \(R_{ij}\) = normalised cross-correlation.
- \(N\) = spreading-code length.
- \(C_i(k)\) = chip value of the first code.
- \(C_j(k)\) = chip value of the second code.

If the correlation is zero, the codes are orthogonal under the alignment being considered.

If the correlation is non-zero, the codes may interfere with each other during despreading.

The absolute value of the correlation indicates the strength of the relationship between the two code sequences.

Lower cross-correlation is generally desirable because it reduces multiple-access interference.

3.10 Bit Error Rate (BER)

Bit Error Rate is an important performance measure used to evaluate the reliability of a digital communication system.

It represents the proportion of transmitted bits that are incorrectly detected at the receiver.

Formula:

\[BER=\frac{\text{Number of incorrectly detected bits}}{\text{Total number of transmitted bits}} \]

For example, if 100,000 bits are transmitted and 100 bits are detected incorrectly:

\[BER=\frac{100}{100000} \]

\[\boxed{BER=0.001} \]

A lower BER indicates better communication performance.

A BER of zero means that no bit errors were observed during the experiment. However, it does not prove that the system will never experience errors. The result depends on the number of bits tested and the simulated channel conditions.

For this assignment, BER should be measured under all three CDMA conditions and for spreading-code lengths of 8, 16 and 64.

3.11 Simulation Implementation Using Python

The following Python program provides a basic DS-CDMA simulation using BPSK modulation and correlation-based detection.

It compares synchronous Walsh codes, random non-orthogonal codes and a near-far condition in which one interfering user has ten times the transmission power and a one-chip timing shift.

The program records the bit error rate and average cross-correlation for different code lengths and numbers of users.

import numpy as npimport csv# Reproducible simulationSEED = 42rng = np.random.default_rng(SEED)CODE_LENGTHS = [8, 16, 64]USER_COUNTS = [2, 4, 8, 16, 32]BITS_PER_USER = 10000EB_N0_DB = 10def hadamard_matrix(n): """Generate a Hadamard matrix of order n.""" H = np.array([[1]]) while H.shape[0] < n: H = np.block([[H, H], [H, -H]]) return Hdef create_walsh_codes(length, users): H = hadamard_matrix(length) return H[:users].astype(float)def create_random_codes(length, users): return rng.choice([-1.0, 1.0],

Important simulation note: This is a simplified teaching model. It uses a one-chip cyclic shift to demonstrate how timing misalignment can destroy the orthogonality of a spreading sequence. The actual near-far behaviour in a real cellular channel also depends on propagation delay, multipath, receiver design and power control. The

simulation's BER values must be obtained by running the code; they should not be replaced with assumed values.

3.12 Recording and Analysing Simulation Results

The simulation should be conducted for code lengths of 8, 16 and 64.

For each code length, different numbers of simultaneous users should be tested without exceeding the number of available codes.

The following table should be completed using the actual output generated by the Python program.

Code length	Number of users	Synchronous Walsh BER	Random-code BER	Near-far BER
8	2	Measured value	Measured value	Measured value
8	4	Measured value	Measured value	Measured value
8	8	Measured value	Measured value	Measured value
16	2	Measured value	Measured value	Measured value
16	4	Measured value	Measured value	Measured value
16	8	Measured value	Measured value	Measured value
16	16	Measured value	Measured value	Measured value
64	2	Measured value	Measured value	Measured value
64	4	Measured value	Measured value	Measured value
64	8	Measured value	Measured value	Measured value

64	16	Measured value	Measured value	Measured value
64	32	Measured value	Measured value	Measured value

The cross-correlation results should also be recorded.

Code length	Code type	Expected correlation behaviour
8	Synchronous Walsh	Zero between different aligned codes
16	Synchronous Walsh	Zero between different aligned codes
64	Synchronous Walsh	Zero between different aligned codes
8	Random binary	Generally non-zero
16	Random binary	Generally non-zero
64	Random binary	Generally non-zero

The actual random-code correlations should be taken from the program output.

3.13 Analysis of the Expected Results

3.13.1 Effect of Increasing the Number of Users

When the number of users increases, more signals share the same frequency spectrum.

In an ideal synchronous system using orthogonal Walsh codes, the receiver can separate the users because the cross-correlation between different aligned codes is zero.

However, when random non-orthogonal codes are used, the signals may interfere with one another. As the number of users increases, the receiver may experience greater multiple-access interference.

Therefore, the BER is expected to increase more noticeably in the random-code condition.

3.13.2 Effect of Increasing the Code Length

Increasing the code length provides a higher processing gain.

For example, a code length of 64 provides 18.06 dB of processing gain, whereas a code length of 8 provides 9.03 dB.

The longer code can provide better separation between users and may reduce the effect of random cross-correlation.

However, a longer spreading code also requires more chips for each information bit. This increases the transmission resources required for each bit.

Therefore, the choice of code length involves a trade-off between processing gain and transmission efficiency.

3.13.3 Effect of Non-Orthogonal Codes

Random non-orthogonal codes can produce interference because their cross-correlation values are not always zero.

When the receiver despreads the received signal using the intended user's code, the signals from other users may produce unwanted components.

This interference can cause incorrect bit decisions.

The effect becomes more important as the number of simultaneous users increases.

3.13.4 Effect of the Near-Far Problem

The near-far problem occurs when one user's signal reaches the receiver at a much higher power level than another user's signal.

For example, a user located close to the base station may produce a strong received signal, while a user located farther away may produce a weak received signal.

If the strong signal has non-zero cross-correlation with the weaker user's spreading code, it can interfere with the detection of the weaker signal.

The receiver may then detect incorrect bits even though the weaker user is using a different spreading code.

The problem becomes more serious when the power difference is large.

3.14 Why Orthogonal Codes May Fail in Practical CDMA Systems

Orthogonal codes provide excellent separation under ideal conditions. However, practical wireless systems may not always maintain perfect orthogonality.

There are several reasons for this.

3.14.1 Lack of Synchronisation

Orthogonal Walsh codes are orthogonal when they are correctly aligned.

If different users' signals arrive at different times, their spreading codes may become misaligned.

This can result in non-zero cross-correlation between signals that were originally orthogonal.

Consequently, the receiver may experience multiple-access interference.

3.14.2 Multipath Propagation

Wireless signals may reach the receiver through multiple propagation paths because of reflections from buildings, structures and other objects.

These reflected signals may arrive with different delays.

The delayed versions of a spreading code may no longer maintain the same orthogonality properties as the original synchronised code.

This can increase interference between users.

3.14.3 Unequal Transmission Powers

When users transmit at significantly different power levels, a strong user's signal may interfere with a weaker user's signal if the receiver experiences imperfect orthogonality.

This is the main reason power control is important in CDMA systems.

3.14.4 Channel Distortion

Wireless channels may introduce noise, fading and phase changes. These effects can reduce the accuracy of signal recovery and increase the probability of bit errors.

Therefore, practical CDMA systems must consider synchronisation, power control and channel conditions rather than relying only on the theoretical properties of spreading codes.

3.15 Importance of Power Control in CDMA

Power control is a technique used to regulate the transmission power of users in a CDMA system.

The objective is to ensure that the received signal powers are maintained at suitable levels.

Without power control, a nearby user may transmit a much stronger signal than a distant user. The strong signal can create interference that affects weaker users.

Power control helps reduce this problem by adjusting the transmission power of each device.

For example:

- If a user's received signal is too strong, the system can instruct the device to reduce its transmission power.

- If a user's received signal is too weak, the system can instruct the device to increase its transmission power, within permitted limits.

Power control improves fairness among users and helps maintain reliable communication.

It also reduces unnecessary transmission power, which can improve battery efficiency in mobile devices.

3.16 Estimating the Number of Simultaneous Users

The number of users that can communicate simultaneously depends on several factors, including:

1. Spreading-code length.
2. Code orthogonality.
3. Signal-to-noise ratio.
4. Transmission power.
5. Synchronisation accuracy.
6. Multipath propagation.
7. Receiver design.
8. Required bit error rate.

In an ideal synchronous Walsh-code system, the maximum number of mutually orthogonal code sequences is equal to the code length.

Therefore:

Code length	Maximum number of orthogonal codes
8	8
16	16
64	64

These are theoretical code-count limits. They do not guarantee that the system can support that many users while meeting a particular BER target under practical channel conditions.

For the purpose of this experiment, a target BER of:

$$\boxed{BER\leq10^{-3}}$$

is selected as an illustrative design requirement.

The simulation should determine the highest number of simultaneous users for which the measured BER remains at or below this target.

The procedure is:

1. Begin with a small number of simultaneous users.
2. Measure BER for each spreading-code length.
3. Increase the number of users gradually.
4. Record the BER after each experiment.
5. Compare the measured BER with the selected target.
6. Identify the maximum tested user count that satisfies the target.

The maximum acceptable number of users should be determined separately for synchronous Walsh codes, random non-orthogonal codes and the near-far condition.

A numerical maximum should only be reported after the simulation has been executed.

3.17 Design Recommendations for a Crowded Stadium

For a crowded stadium environment, CDMA can provide simultaneous communication by allowing multiple users to share the same frequency band through different spreading codes.

However, the system must be designed carefully because the number of users and the radio environment can affect performance.

The following design recommendations are suitable:

1. Use spreading codes with good correlation properties.
2. Maintain accurate synchronisation between users where orthogonal codes are used.
3. Implement power control to reduce the near-far problem.
4. Select an appropriate spreading-code length based on the required number of users.
5. Monitor the received signal quality and bit error rate.
6. Consider multipath propagation and fading in the radio channel.
7. Avoid exceeding the practical user capacity of the system.
8. Use suitable receiver processing to reduce interference.

A longer spreading code can improve processing gain, but the system must balance this advantage against the increased chip transmission requirement.

3.18 Limitations of the Simulation

The simulation provides a simplified representation of a DS-CDMA communication system.

The following limitations should be considered:

1. The simulation uses BPSK modulation and does not represent every practical modulation technique.
2. Walsh-code orthogonality is considered under ideal synchronous conditions in the first experiment.
3. The near-far experiment introduces a simplified one-chip timing shift.
4. Real wireless channels may include multipath propagation, fading and frequency-selective effects.
5. The simulation does not fully represent practical receiver algorithms or channel estimation.
6. The maximum number of supported users depends on the selected BER target and channel conditions.
7. The simulated results should not be treated as guaranteed performance figures for a real cellular network.

These limitations are important because the purpose of the simulation is to demonstrate the fundamental principles of CDMA rather than reproduce every feature of a commercial mobile communication system.

3.19 Conclusion

Code Division Multiple Access allows multiple users to share the same frequency band simultaneously by assigning unique spreading codes.

The spreading process increases the bandwidth of the original information signal, while despreading allows the receiver to recover the intended user's information. Processing gain increases with spreading-code length, providing improved resistance to certain types of interference.

The comparison between orthogonal Walsh codes and random non-orthogonal codes demonstrates the importance of code correlation. Synchronous orthogonal codes can provide strong user separation under ideal conditions, whereas non-orthogonal codes may create additional multiple-access interference.

The near-far problem is another important challenge in CDMA systems. When one user transmits with significantly higher power than another, the stronger signal may interfere with the weaker signal if orthogonality is affected by timing errors or multipath propagation.

Power control, accurate synchronisation and suitable spreading-code selection are therefore essential for reliable CDMA operation.

Overall, CDMA can support multiple simultaneous users, but its performance depends on code length, correlation, transmission power, synchronisation and radio-channel conditions. The simulation provides a method for evaluating these factors through bit error rate and cross-correlation measurements

QUESTION 4 – OFDMA: GIVING EACH USER THE SUBCARRIERS THEY CAN USE BEST

Marks: 6

4.1 Introduction

Orthogonal Frequency Division Multiple Access (OFDMA) is a multiple-access technique used in modern wireless communication systems. It divides the available bandwidth into several closely spaced orthogonal subcarriers and allocates these subcarriers to different users.

OFDMA allows multiple users to communicate simultaneously using the same frequency band. Since the subcarriers are orthogonal to one another, they can overlap in frequency without causing inter-carrier interference under ideal synchronisation conditions.

In this experiment, an OFDMA system consisting of eight users and 64 available subcarriers is considered. The users experience frequency-selective fading, meaning that different subcarriers have different channel conditions.

Three subcarrier-allocation strategies are studied: fixed contiguous allocation, interleaved round-robin allocation and channel-aware best-subcarrier allocation. Their performance is evaluated using throughput and fairness.

4.2 Objectives

1. To understand the working principle of OFDMA.
2. To study the relationship between subcarrier spacing and symbol duration.
3. To understand the purpose of the cyclic prefix.
4. To verify the orthogonality of OFDM subcarriers.
5. To implement different subcarrier-allocation strategies.
6. To compare throughput and fairness among the allocation strategies.
7. To study the effect of frequency-selective fading.
8. To identify a suitable allocation strategy that provides a minimum data rate to every user.

4.3 Working Principle of OFDMA

In OFDMA, the available frequency bandwidth is divided into multiple orthogonal subcarriers. These subcarriers are allocated to different users according to the scheduling requirements.

The transmitter converts the input data into parallel data streams and maps them onto the assigned subcarriers. An Inverse Fast Fourier Transform (IFFT) converts these frequency-domain signals into a time-domain signal for transmission.

At the receiver, a Fast Fourier Transform (FFT) separates the received signal into individual subcarriers, allowing the original data to be recovered.

The system considered in this experiment has the following parameters:

Number of users = 8

Total number of subcarriers = 64

Average subcarriers per user = 64 / 8 = 8

Therefore, under equal allocation, each user receives eight subcarriers.

4.4 Subcarrier Spacing and Symbol Duration

The spacing between adjacent OFDM subcarriers is inversely proportional to the useful OFDM symbol duration.

The relationship is:

$\Delta f = 1 / Tu$

Where:

Δf = Subcarrier spacing in Hz.

Tu = Useful OFDM symbol duration in seconds.

Assume a subcarrier spacing of 15 kHz.

Δf = 15,000 Hz

Tu = 1 / 15,000

Tu = 66.67 μs

Therefore, the useful OFDM symbol duration is approximately 66.67 microseconds.

A smaller subcarrier spacing produces a longer symbol duration, whereas a larger subcarrier spacing produces a shorter symbol duration.

4.5 Cyclic Prefix

A cyclic prefix (CP) is a copy of the ending portion of an OFDM symbol that is added to its beginning before transmission.

It is used to reduce inter-symbol interference caused by multipath propagation. In wireless communication, signals may travel through different paths and arrive at the receiver at different times. These delayed signals can interfere with subsequent OFDM symbols.

The cyclic prefix provides a guard interval between consecutive symbols and helps preserve subcarrier orthogonality when the channel delay spread is within the cyclic prefix duration.

Assume the cyclic prefix duration is 4.69 μs.

Total OFDM symbol duration:

Ts = Tu + TCP

Ts = 66.67 + 4.69

Ts = 71.36 μs

Therefore, the total OFDM symbol duration is approximately 71.36 microseconds.

4.6 Verification of Subcarrier Orthogonality

Orthogonality is an important property of OFDM that allows multiple subcarriers to overlap in frequency without interfering with one another under ideal conditions.

The orthogonality condition for discrete OFDM samples is:

$\Sigma \exp(j2\pi(k-l)n/N) = 0$, for $k \neq l$

Where:

N = Number of samples or subcarriers.

k and l = Subcarrier indices.

For this experiment, N = 64.

Consider three subcarriers with indices 0, 1 and 2.

The inner product between different subcarriers is zero:

C(0,1) = 0

C(0,2) = 0

C(1,2) = 0

The normalised inner product of a subcarrier with itself is equal to 1.

Therefore, the three selected subcarriers are orthogonal to one another.

4.6.1 Python Program for Orthogonality Verification

import numpy as npN = 64subcarriers = [0, 1, 2]signals = {}for k in subcarriers: n = np.arange(N) signals[k] = np.exp(1j * 2 * np.pi * k * n / N)print("Orthogonality Verification")for i in subcarriers: for j in subcarriers: correlation = np.vdot(signals[i], signals[j]) / N print(f"Subcarriers {i} and {j}: {correlation}")

Expected Observation:

The correlation of a subcarrier with itself is approximately 1, while the correlation between two different subcarriers is approximately 0.

This verifies the orthogonality of the selected OFDM subcarriers.

4.7 Frequency-Selective Fading

Frequency-selective fading occurs when different frequency components of a wireless signal experience different levels of attenuation.

In an OFDMA system, different users may experience different channel gains on different subcarriers.

For example, User 1 may experience strong channel conditions on subcarriers 2, 5 and 10, whereas User 2 may experience strong conditions on subcarriers 15, 20 and 25.

Therefore, assigning the same fixed subcarriers to every user may not provide the highest possible throughput.

OFDMA can improve system performance by allocating subcarriers according to the channel conditions of individual users.

4.8 Subcarrier Allocation Strategies

4.8.1 Fixed Contiguous Allocation

In fixed contiguous allocation, the available subcarriers are divided into continuous groups and assigned to different users.

Since there are eight users and 64 subcarriers:

64 / 8 = 8 subcarriers per user.

The allocation is shown below.

User	Assigned Subcarriers
User 1	0–7
User 2	8–15
User 3	16–23
User 4	24–31
User 5	32–39
User 6	40–47
User 7	48–55
User 8	56–63

This method is simple to implement and provides equal frequency resources to all users. However, it does not consider individual channel conditions.

4.8.2 Interleaved Round-Robin Allocation

In interleaved round-robin allocation, subcarriers are distributed among users in a repeating pattern.

For example:

User 1 receives subcarriers 0, 8, 16, 24, 32, 40, 48 and 56.

User 2 receives subcarriers 1, 9, 17, 25, 33, 41, 49 and 57.

User 3 receives subcarriers 2, 10, 18, 26, 34, 42, 50 and 58.

The same pattern continues for the remaining users.

This method distributes frequency resources across the entire bandwidth. It provides equal numbers of subcarriers to users but does not always assign the strongest subcarriers to each user.

4.8.3 Channel-Aware Best-Subcarrier Allocation

In channel-aware allocation, the base station estimates the channel gain of every user on every subcarrier.

Each subcarrier is assigned to the user with the highest channel gain on that subcarrier.

The allocation rule is:

$u^*(k) = \arg\max |h_{u,k}|^2$

Where:

$u^*(k)$ = Selected user for subcarrier k.

$h_{u,k}$ = Channel coefficient of user u on subcarrier k.

$|h_{u,k}|^2$ = Channel power gain.

This method can improve total system throughput because users are assigned subcarriers with favourable channel conditions.

However, users with consistently weak channel conditions may receive fewer resources, resulting in unfairness.

4.9 Throughput Calculation

Throughput represents the amount of successfully transmitted data per unit time.

The theoretical data rate can be estimated using the Shannon capacity formula:

$R = B \log_2(1 + SNR)$

Where:

R = Data rate in bits per second.

B = Bandwidth in Hz.

SNR = Signal-to-noise ratio in linear scale.

For multiple subcarriers, the total system throughput is:

$R_{total} = \Sigma B_{sc} \log_2(1 + SNR_k)$

Where:

B_{sc} = Bandwidth of one subcarrier.

SNRk = Signal-to-noise ratio of the assigned user on subcarrier k.

The actual achievable throughput also depends on modulation, coding, protocol overhead and implementation limitations.

4.10 Fairness Measurement Using Jain's Fairness Index

Throughput alone is not sufficient to evaluate an OFDMA allocation strategy. A system may provide high total throughput while giving very low data rates to some users.

Therefore, fairness is measured using Jain's Fairness Index.

$J = (\Sigma Ri)^2 / [U \times \Sigma Ri^2]$

Where:

J = Jain's Fairness Index.

U = Number of users.

Ri = Throughput of user i.

The value ranges from 1/U to 1.

A value of 1 represents perfect fairness, meaning that all users receive equal throughput. A lower value indicates unequal resource distribution.

4.11 Experimental Design and Simulation Parameters

The experiment considers eight users sharing 64 OFDMA subcarriers.

Frequency-selective fading is generated using Rayleigh fading coefficients. A fixed random seed is used to make the experiment repeatable.

The three allocation strategies are tested under identical channel conditions.

Parameter	Value
Number of users	8
Number of subcarriers	64
Subcarrier spacing	15 kHz
Useful symbol duration	66.67 μs
Cyclic prefix duration	4.69 μs
Total symbol duration	71.36 μs
Channel model	Rayleigh fading

Number of realisations	1,000
Random seed	42
Transmission power	10 (normalised)
Noise power	1 (normalised)

4.11.1 Python Implementation

import numpy as npimport pandas as pdSEED = 42rng = np.random.default_rng(SEED)NUM_SUBCARRIERS = 64NUM_REALISATIONS = 1000SUBCARRIER_SPACING = 15000USEFUL_SYMBOL_TIME = 1 / SUBCARRIER_SPACINGCP_TIME = 4.69e-6SYMBOL_TIME = USEFUL_SYMBOL_TIME + CP_TIMEPOWER = 10NOISE_POWER = 1SUBCARRIER_BANDWIDTH = 15000def jain_fairness(rates): denominator = len(rates) * np.sum(rates ** 2) if denominator == 0: return 0 return (np.sum(rates) ** 2) / denominatordef calculate_rate(gain): snr = POWER * gain / NOISE_POWER rate = SUBCARRIER_BANDWIDTH * np.log2(1 + snr) rate = rate * USEFUL_SYMBOL_TIME / SYMBOL_TIME return rate

Note: This is a simplified simulation using theoretical Shannon capacity and equal power per subcarrier. The channel-aware method does not enforce a minimum-rate requirement, so its fairness may be lower than that of equal allocation.

4.12 Experimental Results and Observations

The following tables must be completed using the actual output generated by the Python program.

Table 4.1: Comparison of Allocation Strategies

Allocation Strategy	Total Throughput (Mbps)	Average User Throughput (Mbps)	Jain Fairness Index
Fixed Contiguous	Measured value	Measured value	Measured value
Interleaved Round Robin	Measured value	Measured value	Measured value
Channel Aware	Measured value	Measured value	Measured value

Table 4.2: Individual User Throughput

User	Fixed Contiguous (Mbps)	Interleaved (Mbps)	Channel Aware (Mbps)
User 1	Measured	Measured	Measured

User 2 Measured	Measured	Measured
User 3 Measured	Measured	Measured
User 4 Measured	Measured	Measured
User 5 Measured	Measured	Measured
User 6 Measured	Measured	Measured
User 7 Measured	Measured	Measured
User 8 Measured	Measured	Measured

Table 4.3: Effect of Increasing the Number of Users

Number of Users	Allocation Strategy	Total Throughput (Mbps)	Jain Fairness Index
2	Fixed Contiguous	Measured	Measured
2	Interleaved	Measured	Measured
2	Channel Aware	Measured	Measured
4	Fixed Contiguous	Measured	Measured
4	Interleaved	Measured	Measured
4	Channel Aware	Measured	Measured
8	Fixed Contiguous	Measured	Measured
8	Interleaved	Measured	Measured
8	Channel Aware	Measured	Measured
16	Fixed Contiguous	Measured	Measured
16	Interleaved	Measured	Measured
16	Channel Aware	Measured	Measured

4.13 Analysis of Throughput and Fairness

4.13.1 Why Can Throughput Maximisation Starve Some Users?

Channel-aware allocation prioritises users with stronger channel conditions.

A user experiencing a strong channel may be selected for several subcarriers because that user can achieve a higher data rate.

However, another user may consistently experience weak channel conditions and receive very few subcarriers.

As a result, total system throughput may increase while some users receive an insufficient data rate.

Therefore, maximising total throughput does not necessarily provide a fair experience to every user.

4.13.2 How Does the Number of Users Affect Channel-Aware Allocation?

When the number of users increases, the probability of finding a user with a strong channel condition on a particular subcarrier also increases.

This phenomenon is known as multiuser diversity.

With more users, the scheduler has more channel conditions to compare and may find a better user for each subcarrier.

Consequently, channel-aware allocation can improve total system throughput.

However, increasing the number of users may also increase unfairness because users with consistently weak channel conditions may receive fewer resources.

4.14 Design Decision: Providing a Minimum Data Rate to Every Student

Consider a lecture hall in which eight students use a wireless network simultaneously. The base station has 64 available subcarriers.

The main requirement is that every student must receive a minimum acceptable data rate.

A system that provides very high throughput to some students while leaving others with extremely low throughput is not suitable for this environment.

A fairness-aware channel-aware allocation strategy is therefore selected.

Initially, each student receives eight subcarriers:

64 / 8 = 8 subcarriers per student.

The scheduler then considers channel conditions and adjusts the remaining resources while maintaining the minimum-rate requirement.

For this experiment, the minimum acceptable rate is defined relative to the equal-allocation reference throughput.

$R_{min} = 0.8 \times R_{ref}$

Where:

Rmin = Minimum required throughput.

Rref = Average per-user throughput under fixed equal allocation.

This is an illustrative experimental requirement rather than a universal wireless communication standard.

The selected strategy is suitable because it considers channel quality while ensuring that every student receives an acceptable level of service.

If the minimum-rate requirement cannot be satisfied for all users, the scheduler should identify the system as overloaded rather than allowing some users to receive inadequate service.

4.15 Limitations of the Experiment

The simulation uses a simplified Rayleigh fading model and theoretical Shannon capacity.

Actual wireless communication systems may experience additional factors such as:

1. Inter-cell interference.
2. Imperfect channel estimation.
3. User mobility.
4. Doppler frequency shifts.
5. Scheduling delays.
6. Channel feedback overhead.
7. Practical modulation and coding limitations.
8. Power-control limitations.
9. Hardware implementation constraints.

Therefore, the simulation provides a useful comparison of allocation strategies but does not represent every aspect of a real commercial OFDMA network.

4.16 Conclusion

This experiment investigated Orthogonal Frequency Division Multiple Access using eight users and 64 available subcarriers.

The relationship between subcarrier spacing and symbol duration was studied, and the importance of the cyclic prefix in reducing inter-symbol interference was explained.

The orthogonality of three OFDM subcarriers was verified mathematically and through a Python simulation.

Three resource-allocation strategies were considered: fixed contiguous allocation, interleaved round-robin allocation and channel-aware best-subcarrier allocation.

Their performance is evaluated using total throughput, individual user throughput and Jain's Fairness Index.

Channel-aware allocation can improve system throughput by assigning subcarriers according to channel conditions. However, it may create unfairness when users experience significantly different channel qualities.

For a lecture hall environment, a fairness-aware channel-aware scheduler is preferred because every student should receive a minimum acceptable data rate. Remaining resources can then be allocated according to channel quality.

Therefore, an effective OFDMA system should balance throughput, fairness and user requirements rather than concentrating only on maximum system capacity.

QUESTION 5 – SDMA: REUSING THE SAME CHANNEL IN DIFFERENT DIRECTIONS

Marks: 6

5.1 Introduction

Space Division Multiple Access (SDMA) is a multiple-access technique that allows multiple users to share the same frequency and time resources by separating them spatially.

Unlike FDMA, which separates users by frequency, and TDMA, which separates users by time slots, SDMA uses the direction or spatial location of users to distinguish their signals.

SDMA commonly uses directional antennas, antenna arrays and beamforming techniques to transmit signals towards specific users while reducing interference in other directions.

In this experiment, a highway cellular communication system is considered. Vehicles travelling in different directions or occupying different angular positions communicate with a base station. The base station uses different antenna configurations to direct signals towards individual vehicles.

The experiment investigates how antenna array size affects beamwidth, sidelobe levels and the ability to separate users spatially.

5.2 Objectives

1. To understand the working principle of SDMA.

2. To study the use of directional antennas in wireless communication.
3. To derive the array factor of a Uniform Linear Array (ULA).
4. To understand the relationship between antenna elements and beamwidth.
5. To compare antenna patterns for different array sizes.
6. To calculate the Signal-to-Interference Ratio (SIR) for users at different angular separations.
7. To determine the minimum angular separation required for two users to share the same frequency and time resource.
8. To evaluate the suitability of SDMA for highway and toll-plaza communication systems.

5.3 Working Principle of SDMA

SDMA allows multiple users to use the same frequency and time resources by assigning different spatial directions to their signals.

A base station equipped with directional antennas can transmit energy towards a particular user while reducing the signal strength in other directions.

For example, consider two vehicles located at different angular positions relative to a base station. The base station can direct one beam towards the first vehicle and another beam towards the second vehicle.

Both vehicles may use the same frequency channel and time slot because their signals are separated spatially.

The main advantage of SDMA is that it improves spectrum utilisation by allowing frequency reuse within the same cell.

However, the effectiveness of SDMA depends on antenna characteristics, angular separation, propagation conditions and interference levels.

5.4 Antenna Configurations

Three antenna configurations are considered in this experiment.

5.4.1 Omnidirectional Antenna

An omnidirectional antenna radiates energy approximately equally in all horizontal directions.

It does not focus its transmission towards a particular user.

Advantages:

1. Provides coverage in multiple directions.
2. Has a simple implementation.
3. Suitable for systems where users are distributed around the base station.

Disadvantages:

1. Provides limited spatial separation.
2. May produce interference between users.
3. Does not concentrate transmission energy towards individual users.

5.4.2 120-Degree Sector Antenna

A sector antenna covers a specific angular region rather than the complete 360-degree area.

A base station can use three 120-degree sectors to provide coverage around the cell.

Each sector serves users within its coverage region.

Advantages:

1. Provides better directional coverage than an omnidirectional antenna.
2. Reduces interference from users located outside the sector.
3. Improves frequency reuse opportunities.

Disadvantages:

1. Requires multiple antenna sectors.
2. Users near sector boundaries may experience interference.
3. Provides less precise spatial separation than adaptive beamforming.

5.4.3 Uniform Linear Array

A Uniform Linear Array (ULA) consists of multiple antenna elements arranged in a straight line with equal spacing.

The signals transmitted by the individual antenna elements are controlled in phase so that their combined radiation pattern forms a beam in a selected direction.

By changing the phase difference between elements, the main beam can be electronically steered towards different users.

A larger antenna array generally provides a narrower main beam, improving the ability to separate users located at different angular positions.

5.5 Derivation of the Array Factor for a Uniform Linear Array

Consider a Uniform Linear Array containing N antenna elements.

Let:

N = Number of antenna elements.

d = Distance between adjacent antenna elements.

λ = Wavelength of the transmitted signal.

θ = Observation angle measured from the array broadside direction.

β = Progressive phase shift between adjacent antenna elements.

The phase difference between adjacent elements is:

$\psi = (2\pi d/\lambda) \sin\theta + \beta$

The array factor is:

$AF(\theta) = \Sigma \exp(jn\psi)$, for n = 0 to N-1

Using the geometric series formula:

$AF(\theta) = [1 - \exp(jN\psi)] / [1 - \exp(j\psi)]$

The normalised magnitude of the array factor is:

$|AF(\theta)| = |\sin(N\psi/2) / \sin(\psi/2)|$

The normalised array factor is:

$AFnorm(\theta) = |\sin(N\psi/2) / [N \sin(\psi/2)]|$

The value of the array factor represents the relative radiation strength in a particular direction.

For beam steering towards an angle θ_0, the phase shift is selected as:

$\beta = -(2\pi d/\lambda) \sin\theta_0$

This makes the antenna elements combine constructively in the desired direction.

For this experiment, the antenna spacing is selected as:

$d = \lambda/2$

This spacing is commonly used in antenna arrays because it helps avoid unwanted grating lobes over the visible angular region.

5.6 Effect of the Number of Antenna Elements

The number of antenna elements has a significant effect on the antenna radiation pattern.

When the number of antenna elements increases:

1. The main beam becomes narrower.
2. Directional gain increases.
3. Spatial separation between users becomes easier.
4. The antenna can concentrate more energy towards a selected direction.
5. The system may distinguish users with smaller angular separation.

For a uniformly weighted broadside linear array with half-wavelength spacing, an approximate half-power beamwidth is:

$HPBW \approx 101.6° / N$

This is an approximate large-array relationship and is less accurate for very small arrays.

The estimated beamwidths are:

Number of Elements	Approximate HPBW
2	50.8°
4	25.4°
8	12.7°
16	6.35°

The sidelobe level of a uniformly weighted array is approximately -13.26 dB for the conventional first sidelobe.

Increasing the number of elements narrows the main beam, but it does not automatically reduce the relative first sidelobe level when uniform amplitude weighting is used.

Additional amplitude weighting techniques, such as Taylor or Dolph-Chebyshev weighting, can reduce sidelobes at the cost of a wider main beam.

5.7 Signal-to-Interference Ratio

Signal-to-Interference Ratio (SIR) measures the strength of the desired signal relative to interfering signals.

It is defined as:

SIR = Desired Signal Power / Interference Power

In decibels:

$SIR(dB) = 10 \log_{10}(\text{Desired Signal Power / Interference Power})$

For this experiment, assume that the base station directs its beam towards User A. User B is located at an angular separation $\Delta\theta$ from User A.

Assuming equal received reference powers and considering the antenna pattern as the main source of spatial discrimination:

$SIR(dB) = -10 \log_{10}(GB)$

Where:

GB = Normalised antenna power gain in the direction of User B.

For this experiment, the required SIR threshold is selected as:

SIRmin = 10 dB

This means that the desired signal should be at least ten times stronger than the interfering signal.

The minimum angular separation is the smallest separation at which the calculated SIR reaches or exceeds the selected threshold.

5.8 Experimental Design

The experiment considers a base station using a Uniform Linear Array with different numbers of antenna elements.

The following configurations are evaluated:

1. Two antenna elements.
2. Four antenna elements.
3. Eight antenna elements.
4. Sixteen antenna elements.

The beam is initially directed towards User A at the broadside direction.

User B is placed at different angular separations from User A, ranging from 5 degrees to 60 degrees.

The antenna gain towards User B is calculated for each separation.

The SIR is then evaluated to determine whether the two users can share the same frequency and time resource.

5.9 Simulation Parameters

Parameter	Value
Antenna type	Uniform Linear Array
Antenna spacing	λ/2
Array configurations	2, 4, 8 and 16 elements
Beam steering direction	0°
Angular separation	5° to 60°
Angular step	1°
Required SIR	10 dB
Antenna weighting	Uniform
Reference signal power	1 (normalised)

5.10 Python Implementation

The following Python program calculates the antenna array factor for different array sizes.

It also calculates the approximate half-power beamwidth, first sidelobe level and minimum angular separation that satisfies the selected SIR threshold.

import numpy as npimport pandas as pdimport matplotlib.pyplot as plt# Simulation parametersELEMENT_COUNTS = [2, 4, 8, 16]D_OVER_LAMBDA = 0.5SIR_THRESHOLD_DB = 10ANGLES = np.linspace(-90, 90, 18001)SEPARATIONS = np.arange(5, 61, 1)# Calculate normalised array factordef array_factor(N, angles): theta = np.radians(angles) beta = 0 psi = (2 * np.pi * D_OVER_LAMBDA * np.sin(theta) + beta) denominator = np.sin(psi / 2) numerator = np.sin(N * psi / 2) af = np.ones_like(theta)

Note: The simulation uses an ideal array model with uniform element weighting and a broadside beam. The sidelobe calculation uses an approximate angular exclusion region and should be checked against the plotted radiation pattern. The SIR calculation assumes equal reference signal powers and ignores other propagation effects.

5.11 Experimental Results and Observations

After executing the Python program, the generated CSV file contains the calculated antenna performance measurements.

The following table should be completed using the actual simulation output.

Table 5.1: Antenna Performance Comparison

Number of Elements	Beamwidth (Degrees)	Sidelobe Level (dB)	Minimum Angular Separation (Degrees)
2	Measured	Measured	Measured
4	Measured	Measured	Measured
8	Measured	Measured	Measured
16	Measured	Measured	Measured

Table 5.2: SIR at Different Angular Separations

Angular Separation	SIR for 2 Elements (dB)	SIR for 4 Elements (dB)	SIR for 8 Elements (dB)	SIR for 16 Elements (dB)
5°	Measured	Measured	Measured	Measured
10°	Measured	Measured	Measured	Measured
15°	Measured	Measured	Measured	Measured
20°	Measured	Measured	Measured	Measured
30°	Measured	Measured	Measured	Measured
40°	Measured	Measured	Measured	Measured
50°	Measured	Measured	Measured	Measured
60°	Measured	Measured	Measured	Measured

5.11.1 Observation

The antenna radiation patterns show how the number of antenna elements affects the directional characteristics of the system.

A larger antenna array generally produces a narrower main beam, allowing the base station to concentrate transmission energy towards a particular user.

When the angular separation between two users increases, the interfering user's position may fall into a lower-gain region of the antenna pattern. This improves the SIR and makes spatial reuse more practical.

However, the exact minimum separation depends on the antenna pattern and the selected SIR threshold.

5.12 Analysis of Practical Limitations

5.12.1 Practical Limits of SDMA

Although SDMA provides efficient spatial reuse, its performance is affected by several practical limitations.

Angular Spread: Signals may arrive from multiple directions because of reflections and scattering. This can make it difficult to distinguish users based only on their direct arrival angles.

Clustered Users: When multiple users are located very close to one another in the same direction, their signals may have similar spatial characteristics.

Antenna Imperfections: Real antenna arrays may experience phase errors, amplitude variations and mutual coupling, which can affect beamforming accuracy.

User Mobility: Moving vehicles change their angular positions relative to the base station. The beamforming system must continuously update its beam direction.

Interference: Other base stations and nearby communication systems may produce interference that cannot be eliminated by directional beamforming alone.

5.12.2 Why Cannot Users in Nearly the Same Direction Be Separated Easily?

Users located in nearly the same direction produce signals with similar spatial characteristics at the base station.

The antenna array may not have sufficient angular resolution to distinguish these signals.

If two users are separated by an angle smaller than the effective beamwidth, their signals may fall within the same main beam.

Consequently, the base station may not be able to suppress one user's signal without also affecting the other user's signal.

Increasing the number of antenna elements can improve angular resolution, but it cannot eliminate all limitations caused by propagation, antenna imperfections and closely clustered users.

5.13 Design Decision: SDMA at a Toll Plaza

Consider a toll plaza where multiple vehicles arrive from the same direction.

Vehicles may be positioned close to one another, particularly during heavy traffic conditions.

Using SDMA alone may not be the most reliable solution because the angular separation between vehicles can be very small.

Even if the base station uses a large antenna array, vehicles travelling in nearly the same direction may remain difficult to separate spatially.

Therefore, SDMA should not be the only multiple-access technique used in this situation.

A hybrid approach combining SDMA with TDMA or OFDMA is more suitable.

SDMA can separate vehicles that are sufficiently far apart in angular direction, while TDMA or OFDMA can allocate separate resources to vehicles that cannot be distinguished spatially.

This approach improves resource utilisation while reducing interference between closely positioned vehicles.

5.14 Conclusion

This experiment investigated Space Division Multiple Access and its application in wireless communication systems.

The working principle of SDMA was studied using omnidirectional antennas, sector antennas and Uniform Linear Arrays.

The array factor of a ULA was derived, and the effect of increasing the number of antenna elements on beamwidth and sidelobe characteristics was explained.

A Python simulation was designed to evaluate antenna patterns for arrays containing 2, 4, 8 and 16 elements.

The experiment also considered the SIR between two users at different angular separations and examined the minimum separation required to satisfy a selected SIR threshold.

The results demonstrate that increasing the number of antenna elements generally produces narrower beams and improves spatial separation.

However, SDMA has practical limitations when users are located close together or experience similar propagation conditions.

For a toll-plaza environment, combining SDMA with TDMA or OFDMA is a more reliable approach than depending entirely on spatial separation.

Therefore, SDMA can significantly improve spectrum utilisation, but its effectiveness depends on antenna design, user distribution and interference conditions.

QUESTION 6 – CHOOSING MULTIPLE-ACCESS TECHNIQUES FOR REAL DEPLOYMENTS

Marks: 7

6.1 Introduction

Multiple-access techniques allow several users to share the available communication resources in a wireless network. Different techniques use different methods to separate users and manage communication resources.

The major multiple-access techniques considered in this assignment are:

1. Space Division Multiple Access (SDMA).
2. Time Division Multiple Access (TDMA).
3. Frequency Division Multiple Access (FDMA).
4. Code Division Multiple Access (CDMA).
5. Orthogonal Frequency Division Multiple Access (OFDMA).

Each technique has different advantages and limitations depending on the communication environment, traffic requirements, interference conditions and number of users.

In this experiment, four real-world wireless communication deployments are considered. The five multiple-access techniques are evaluated using weighted selection criteria to identify the most suitable technique for each deployment.

The results and observations from Questions 2, 3, 4 and 5 are used to support the final selection.

6.2 Objectives

1. To understand the characteristics of different multiple-access techniques.
2. To identify important criteria for selecting a multiple-access technique.
3. To assign weights to the selection criteria according to their importance.
4. To compare SDMA, TDMA, FDMA, CDMA and OFDMA.
5. To select suitable techniques for four real-world communication deployments.
6. To evaluate hybrid multiple-access approaches.
7. To perform sensitivity analysis by changing the criterion weights.
8. To identify the least suitable technique for different deployment requirements.

6.3 Multiple-Access Techniques Considered

6.3.1 Space Division Multiple Access (SDMA)

SDMA separates users according to their spatial locations or directions.

It uses directional antennas and beamforming to transmit signals towards specific users while reducing interference in other directions.

The main advantage of SDMA is that multiple users can reuse the same frequency and time resources if they are sufficiently separated spatially.

However, SDMA may not perform effectively when users are located very close together or have similar spatial characteristics.

6.3.2 Time Division Multiple Access (TDMA)

TDMA divides the available communication channel into different time slots.

Each user receives a specific time slot during which communication takes place.

TDMA is relatively simple to implement and provides predictable resource allocation.

However, synchronisation and time-slot management are required. Dedicated time slots may also be wasted when users have no data to transmit.

6.3.3 Frequency Division Multiple Access (FDMA)

FDMA divides the available frequency bandwidth into separate frequency channels.

Each user receives a dedicated frequency channel for communication.

FDMA is suitable for continuous communication because users can transmit without waiting for another user's time slot.

However, frequency guard bands reduce the available bandwidth, and dedicated channels may remain unused during periods of low traffic.

6.3.4 Code Division Multiple Access (CDMA)

CDMA allows multiple users to share the same frequency band and time interval using different spreading codes.

The receiver uses the corresponding code to separate the desired user's signal from other users.

CDMA provides interference tolerance and can support multiple simultaneous users.

However, performance depends on code correlation, synchronisation and power control. The near-far problem can significantly affect communication when users arrive at the receiver with unequal signal strengths.

6.3.5 Orthogonal Frequency Division Multiple Access (OFDMA)

OFDMA divides the available bandwidth into orthogonal subcarriers and allocates different groups of subcarriers to different users.

It supports flexible resource allocation and can adapt to frequency-selective fading.

OFDMA is particularly suitable for high-data-rate wireless networks with varying traffic requirements.

However, it requires channel-state information and scheduling. Poor scheduling decisions may also result in unfair resource distribution.

6.4 Deployment Scenarios

Four real-world communication deployments are considered.

6.4.1 Wide-Area Cellular Voice Communication

This deployment provides voice communication over a large geographical area.

Users may move between different cells while maintaining active calls.

The system requires reliable communication, efficient frequency reuse, mobility support and acceptable interference levels.

The main requirements are:

1. Continuous voice communication.
2. Reliable coverage over a wide area.

3. Support for user mobility.
4. Efficient spectrum utilisation.
5. Low call interruption during handover.

6.4.2 Jam-Resistant Tactical Communication Link

This deployment is used for tactical communication in an environment where intentional or unintentional interference may occur.

The communication system must maintain connectivity under difficult radio conditions.

The main requirements are:

1. Strong interference tolerance.
2. Reliable communication.
3. Efficient use of available spectrum.
4. Ability to operate under changing radio conditions.
5. Robustness against signal interference and fading.

No single multiple-access technique guarantees resistance to deliberate jamming. Actual performance also depends on frequency planning, coding, power control, antenna design and other communication-security measures.

6.4.3 High-Rate Stadium Downlink

This deployment provides high-speed wireless connectivity to a large number of users inside a stadium.

Many users may access the network simultaneously for video streaming, social media, live updates and file downloads.

The main requirements are:

1. High total data throughput.
2. Support for a large number of users.
3. Efficient bandwidth utilisation.
4. Adaptation to changing traffic demand.
5. Interference management in a crowded environment.

6.4.4 Low-Rate Sensor Network

This deployment consists of approximately 5,000 sensors transmitting small reports periodically.

Each sensor sends a small amount of data to a central monitoring system.

The main requirements are:

1. Support for a large number of devices.
2. Low communication overhead.
3. Efficient resource utilisation.
4. Low energy consumption.
5. Reliable delivery of sensor reports.

6.5 Weighted Selection Criteria

To compare the five multiple-access techniques, five important selection criteria are defined.

Each criterion is assigned a weight according to its importance in the overall deployment.

The weights add up to 100%.

6.5.1 Selection Criteria

1. Spectral Efficiency

Spectral efficiency measures how effectively the available frequency bandwidth is used to transmit data.

A technique with higher spectral efficiency can support more communication traffic using the same bandwidth.

2. Interference and Robustness

This criterion measures the ability of a technique to maintain communication under interference, fading and other radio disturbances.

3. Scalability

Scalability measures how effectively the technique supports an increasing number of users or devices.

4. Traffic Flexibility

Traffic flexibility measures how effectively a technique handles different traffic patterns, including continuous voice, bursty data and periodic sensor reports.

5. Implementation Complexity

This criterion measures the complexity of implementing and managing the multiple-access technique.

A technique with lower implementation complexity generally requires fewer advanced scheduling and signal-processing operations.

6.6 Weight Assignment for Different Deployments

Different deployments have different priorities. Therefore, the selection criteria are assigned different weights for each scenario.

Table 6.1: Weighted Selection Criteria

Selection Criterion	Cellular Voice	Tactical Link	Stadium Downlink	Sensor Network
Spectral Efficiency	25%	15%	30%	20%
Interference and Robustness	25%	40%	20%	20%
Scalability	20%	15%	20%	25%
Traffic Flexibility	20%	15%	20%	15%
Implementation Complexity	10%	15%	10%	20%
Total	100%	100%	100%	100%

6.6.1 Justification of Weights

For wide-area cellular voice communication, spectral efficiency and interference management receive high weights because the system must support many users while maintaining reliable calls.

For tactical communication, interference robustness receives the highest weight because maintaining connectivity under difficult radio conditions is essential.

For a stadium downlink, spectral efficiency receives the highest weight because a large number of users require high data rates simultaneously.

For a sensor network, scalability receives the highest weight because the system must support thousands of devices. Implementation complexity and energy-related considerations are also important.

The weights represent the priorities of the selected deployment scenarios and are not universal industry standards.

6.7 Rating Scale

Each multiple-access technique is rated using a scale from 1 to 5.

Rating	Meaning
1	Very Poor
2	Poor
3	Average
4	Good
5	Excellent

A higher rating indicates better suitability for the selected criterion.

The weighted score is calculated using:

Weighted Score = Σ (Criterion Weight × Rating)

For convenience, the final score is converted to a scale of 100:

Final Score = Σ (Weight Percentage × Rating) / 5

The maximum possible final score is 100.

The ratings are engineering judgements based on the characteristics of the techniques and the earlier experiments. They are illustrative comparative ratings rather than measured benchmark results.

6.8 Comparison of Multiple-Access Techniques

The following general comparison is used as the basis for the deployment-specific ratings.

Criterion	SDMA	TDMA	FDMA	CDMA	OFDMA
Spectral Efficiency	High with effective beamforming	Moderate	Moderate	Good under controlled conditions	High
Interference Robustness	Depends on spatial separation	Depends on time synchronisation	Depends on frequency separation	Good with effective power control	Good with scheduling and interference management

Scalability	Depends on antenna capacity	Limited by available slots	Limited by available channels	Limited by interference and code capacity	Flexible resource allocation
Traffic Flexibility	Good with adaptive beamforming	Moderate	Limited with fixed channels	Good for shared access	High
Implementation Complexity	High	Low to Moderate	Low	High	High

6.9 Weighted Evaluation for Wide-Area Cellular Voice

For wide-area cellular voice communication, reliability, mobility support and spectrum utilisation are important.

The following ratings are assigned.

Table 6.2: Cellular Voice Evaluation

Technique	Spectral Efficiency (25%)	Robustness (25%)	Scalability (20%)	Flexibility (20%)	Complexity (10%)	Final Score
SDMA	4	3	4	4	2	70
TDMA	3	3	3	3	5	64
FDMA	2	3	2	2	5	51
CDMA	4	4	4	4	2	74
OFDMA	5	4	5	5	2	88

6.9.1 Selected Technique

Selected technique: OFDMA

OFDMA receives the highest illustrative weighted score because it provides flexible resource allocation, high spectral efficiency and support for multiple users.

It can allocate different subcarriers to users according to their channel conditions and traffic requirements.

However, a practical cellular voice network may combine multiple-access methods. For example, frequency bands may be divided using FDMA while users are scheduled over OFDMA resources within each band.

The final choice therefore represents the preferred radio-access method for the evaluated scenario rather than a claim that cellular systems use OFDMA alone.

6.10 Weighted Evaluation for Jam-Resistant Tactical Communication

For tactical communication, interference robustness receives the highest weight.

The system must maintain communication under changing radio conditions and interference.

Table 6.3: Tactical Communication Evaluation

Technique	Spectral Efficiency (15%)	Robustness (40%)	Scalability (15%)	Flexibility (15%)	Complexity (15%)	Final Score
SDMA	4	3	3	4	2	62
TDMA	3	3	3	3	5	63
FDMA	2	3	2	2	5	55
CDMA	4	5	4	4	2	84
OFDMA	5	3	5	5	2	74

6.10.1 Selected Technique

Selected technique: CDMA

CDMA is selected because it uses spreading codes to allow multiple users to share the same frequency band.

Spread-spectrum communication can provide processing gain and improve tolerance to certain forms of interference.

However, CDMA performance depends on code properties, synchronisation and power control.

As demonstrated in Question 3, the near-far problem can increase interference when users arrive at the receiver with significantly different signal strengths.

Therefore, CDMA should be combined with suitable power control, coding and other interference-management mechanisms.

The selection does not imply that CDMA is immune to deliberate jamming. A real tactical system would require additional anti-interference measures.

6.11 Weighted Evaluation for High-Rate Stadium Downlink

A stadium network must support a large number of users simultaneously.

High throughput and efficient bandwidth utilisation are the main requirements.

Table 6.4: Stadium Downlink Evaluation

Technique	Spectral Efficiency (30%)	Robustness (20%)	Scalability (20%)	Flexibility (20%)	Complexity (10%)	Final Score
SDMA	5	4	4	4	2	82
TDMA	3	3	3	3	5	64
FDMA	2	3	2	2	5	51
CDMA	4	3	3	4	2	66
OFDMA	5	4	5	5	2	88

6.11.1 Selected Technique

Selected technique: OFDMA

OFDMA is selected because it supports high data rates and flexible resource allocation.

The base station can assign different groups of subcarriers to different users according to channel quality and traffic demand.

The observations from Question 4 show that channel-aware allocation can improve overall throughput.

However, fairness must also be considered because users with weaker channel conditions may receive fewer resources.

For a stadium network, OFDMA can be combined with SDMA using sector antennas or beamforming to improve spatial reuse and serve users in different areas.

6.12 Weighted Evaluation for a Large Sensor Network

The sensor network contains approximately 5,000 devices that periodically transmit small reports.

The system must support a large number of devices while avoiding unnecessary resource reservation.

Table 6.5: Sensor Network Evaluation

Technique	Spectral Efficiency (20%)	Robustness (20%)	Scalability (25%)	Flexibility (15%)	Complexity (20%)	Final Score
SDMA	4	3	3	3	2	61
TDMA	3	3	3	3	5	68
FDMA	2	3	2	2	5	55
CDMA	4	4	4	4	2	72
OFDMA	5	4	5	5	2	86

6.12.1 Selected Technique

Selected technique: OFDMA with dynamic scheduling

OFDMA is selected because it can allocate resources according to the actual traffic requirements of the sensor devices.

The sensors transmit small reports periodically, meaning that they do not require continuous dedicated frequency channels.

A dynamic scheduler can allocate subcarriers when devices have data to transmit.

This avoids wasting resources on permanently reserved channels.

For very low-power sensor networks, a dedicated low-power communication protocol may be more suitable than a conventional high-throughput OFDMA system. Therefore, the selected technique represents a general resource-allocation approach rather than a complete sensor-network protocol design.

6.13 Summary of Final Selections

Table 6.6: Recommended Multiple-Access Techniques

Deployment	Selected Technique	Main Reason
Wide-Area Cellular Voice	OFDMA	Flexible scheduling and high spectral efficiency
Jam-Resistant Tactical Link	CDMA	Spread-spectrum interference tolerance
High-Rate Stadium Downlink	OFDMA with SDMA	High throughput and spatial reuse

Large Sensor Network	OFDMA with Dynamic Scheduling	Flexible allocation for periodic bursty traffic

6.14 Hybrid Multiple-Access Techniques

In practical wireless communication systems, multiple-access techniques are often combined rather than used independently.

A hybrid system combines the advantages of different techniques to improve overall communication performance.

6.14.1 FDMA Combined with TDMA

FDMA divides the available spectrum into different frequency bands, while TDMA divides each frequency band into time slots.

This combination allows multiple users to share a frequency band using separate time slots.

It provides predictable resource allocation and reduces the need for every user to occupy a separate frequency channel.

This combination has historically been used in GSM cellular systems.

6.14.2 OFDMA Combined with SDMA

OFDMA allocates different subcarriers to users, while SDMA separates users according to their spatial directions.

A base station can use beamforming to serve users in different directions while allocating subcarriers according to their channel conditions.

This combination can improve spectrum utilisation in crowded environments such as stadiums.

6.14.3 CDMA Combined with Power Control

CDMA systems benefit from power control because users transmit signals with different received power levels.

Without suitable power control, a nearby user may produce a much stronger signal than a distant user.

This can create the near-far problem and increase interference.

Power control helps maintain more balanced received signal strengths.

6.15 Sensitivity Analysis

Sensitivity analysis is performed to understand how changes in selection-criterion weights affect the final technique selection.

The original weights are changed by approximately 20%, and the weighted scores are recalculated after normalising the adjusted weights to a total of 100%.

The purpose is to determine whether the selected technique remains suitable when the importance of individual criteria changes.

For example, in the stadium deployment, spectral efficiency receives the highest original weight.

If its weight is reduced and scalability or interference robustness receives greater importance, the final scores may change.

Similarly, the tactical communication selection may change if implementation simplicity becomes more important than interference robustness.

6.15.1 Sensitivity Analysis Procedure

1. Record the original selection-criterion weights.
2. Increase one criterion weight by 20%.
3. Decrease the same criterion weight by 20%.
4. Adjust the remaining weights proportionally so that the total remains 100%.
5. Recalculate the weighted scores.
6. Compare the revised scores with the original scores.
7. Identify whether the selected technique changes.

The sensitivity analysis should be performed separately for all four deployments.

Table 6.7: Sensitivity Analysis Record

Deployment	Original Selected Technique	Result After +20% Weight Change	Result After -20% Weight Change	Selection Stable?
Cellular Voice	OFDMA	Enter measured result	Enter measured result	Yes/No
Tactical Link	CDMA	Enter measured result	Enter measured result	Yes/No
Stadium Downlink	OFDMA	Enter measured result	Enter measured result	Yes/No

Sensor Network	OFDMA	Enter measured result	Enter measured result	Yes/No

The final stability of each selection must be determined from the recalculated scores. It should not be assumed without performing the calculation.

6.16 Why Do Real Wireless Systems Combine Multiple-Access Techniques?

Real wireless communication systems combine different techniques because no single method provides the best performance for every requirement.

For example, FDMA and TDMA can be combined to divide spectrum into frequency channels and then allocate time slots within those channels.

Similarly, OFDMA and SDMA can be combined to improve both frequency-resource allocation and spatial reuse.

Hybrid systems provide several advantages:

1. Improved spectrum utilisation.
2. Better interference management.
3. Flexible resource allocation.
4. Support for different traffic patterns.
5. Improved scalability.
6. Better adaptation to changing channel conditions.

The use of hybrid techniques allows communication systems to balance capacity, reliability, complexity and fairness.

6.17 Least Robust Multiple-Access Technique

The least suitable technique depends on the deployment requirements.

For a high-rate stadium downlink, FDMA with permanently dedicated frequency channels may be less suitable because it can waste bandwidth when users have different traffic requirements.

For a crowded sensor network, fixed dedicated TDMA or FDMA allocation may also be inefficient because most sensors transmit only small amounts of data periodically.

For a tactical communication environment, a technique without effective interference-management mechanisms may fail to maintain reliable communication under strong interference.

Therefore, no single technique can be identified as the least robust in every environment. Its suitability depends on traffic patterns, interference conditions, user distribution and system requirements.

6.18 Overall Conclusion

This experiment evaluated five multiple-access techniques: SDMA, TDMA, FDMA, CDMA and OFDMA.

Four real-world communication deployments were considered: wide-area cellular voice, jam-resistant tactical communication, high-rate stadium downlink and a large low-rate sensor network.

Five selection criteria were defined and assigned different weights according to the requirements of each deployment.

The weighted evaluation indicated that OFDMA is suitable for applications requiring high throughput and flexible resource allocation. CDMA is a possible choice for tactical communication where spread-spectrum interference tolerance is important, although additional protection against jamming is necessary.

The experiment also demonstrated that hybrid techniques can provide better performance than relying on a single multiple-access method.

The sensitivity analysis provides a way to examine whether the selected technique remains appropriate when deployment priorities change.

Overall, the selection of a multiple-access technique should consider spectral efficiency, interference robustness, scalability, traffic flexibility and implementation complexity. The best solution is the one that satisfies the actual requirements of the deployment rather than simply achieving the highest score in one performance category.

QUESTION 7 – DIAGNOSE, REDESIGN AND VALIDATE (13 MARKS)

7.1 Introduction

The experiments conducted in Questions 1 to 6 demonstrate that different multiple-access techniques have different advantages and limitations. A technique that performs well under one set of conditions may become inefficient when the number of users increases or when operating conditions change.

In this question, the limitations of an existing solution are examined, a redesigned strategy is proposed, and the improved strategy is tested under the same experimental conditions.

The selected problem is the fairness limitation of channel-aware OFDMA allocation identified in Question 4.

In channel-aware OFDMA, subcarriers are assigned to users with the best channel conditions. This improves total system throughput, but users experiencing poor channel conditions may receive fewer subcarriers or lower data rates. Therefore, maximising total throughput does not necessarily provide fair service to every user.

7.2 Selected Limitation

The selected limitation is the possibility of user starvation in channel-aware OFDMA allocation.

In a frequency-selective wireless environment, different users experience different channel gains across the available subcarriers. A channel-aware scheduler assigns each subcarrier to the user with the highest channel gain on that subcarrier.

This approach has the following advantages:

1. It uses subcarriers efficiently.
2. It increases the total system throughput.
3. It takes advantage of favourable channel conditions.
4. It reduces the allocation of resources to users with poor channel quality.

However, it also has some disadvantages:

1. Users with consistently weak channel conditions may receive fewer resources.
2. Users with favourable channel conditions may receive most of the available subcarriers.
3. Some users may experience very low throughput.
4. The system may become unfair even when its overall throughput is high.

Therefore, the original channel-aware allocation method needs to be improved to provide a better balance between throughput and fairness.

7.3 Diagnosis of the Bottleneck

The main bottleneck is the allocation strategy used by the OFDMA scheduler.

The original scheduler selects the user with the highest channel gain for each subcarrier. It does not consider how much data each user has already received.

For example, consider a system containing eight users and sixty-four subcarriers. If two users consistently experience better channel conditions, they may receive a larger share of the available subcarriers.

The remaining users may receive fewer resources and experience lower throughput.

The bottleneck can be diagnosed using the following measurable quantities:

- Total system throughput.
- Individual user throughput.
- Minimum user throughput.
- Maximum user throughput.
- Jain's fairness index.
- Percentage of users receiving a reasonable share of system resources.

The Jain's fairness index is calculated as:

$$J=\frac{(\sum_{i=1}^{N}R_i)^2}{N\sum_{i=1}^{N}R_i^2}$$

Where:

- J = Jain's fairness index.
- N = Number of users.
- R_i = Throughput of user i.

The fairness index ranges from $1/N$ to 1.

A value close to 1 indicates that throughput is distributed more equally among users. A lower value indicates that some users receive significantly more throughput than others.

7.4 Measurable Objective

The objective of the redesign is to improve fairness without reducing total system throughput excessively.

The following performance targets are selected:

1. Achieve a Jain's fairness index of at least 0.95.
2. Maintain at least 80% of the original channel-aware allocation's average total throughput.
3. Ensure that every user receives at least one subcarrier.
4. Evaluate the performance over at least 1,000 independent channel realisations.
5. Compare both allocation methods using identical channel conditions.

These targets are selected as experimental design requirements. Whether they are achieved must be verified using the simulation results.

7.5 Proposed Redesigned Strategy

The proposed solution is a fairness-aware OFDMA allocation strategy.

Instead of assigning every subcarrier to the user with the highest channel gain, the redesigned scheduler considers the throughput already received by each user.

The scheduler uses a max-min throughput allocation approach.

The working procedure is:

1. Initially, all users have zero allocated throughput.
2. The scheduler identifies the user currently receiving the lowest throughput.
3. It assigns that user the best available unallocated subcarrier.
4. The user's throughput is updated.
5. The scheduler repeats the process until all subcarriers are allocated.
6. The final throughput and fairness values are calculated.

This approach attempts to improve the service received by users with lower throughput while still selecting favourable subcarriers for individual users.

The redesigned strategy is expected to improve fairness compared with the original maximum-channel-gain allocation method.

7.6 Experimental Setup

The simulation uses the following parameters.

Parameter	Value
Number of users	8
Number of subcarriers	64
Subcarrier spacing	15 kHz
Channel model	Rayleigh fading
Number of channel realisations	1,000
Random seed	42
Noise power	1 (normalised)

Transmission power per subcarrier	10 (normalised)
Original allocation	Maximum channel gain
Redesigned allocation	Fairness-aware max-min
Throughput measurement	Mbps-equivalent normalised rate

Both allocation methods use the same generated channel conditions. This ensures that the comparison is fair and that any performance difference is caused by the allocation strategy rather than different channel samples.

7.7 Implementation Using Python

The following Python program compares the original channel-aware OFDMA allocation with the redesigned fairness-aware allocation.

The program generates Rayleigh fading channel gains, calculates the achievable rate of each user on each subcarrier, performs both allocation methods, and measures throughput and fairness.

import numpy as npimport pandas as pd# Simulation parametersNUM_USERS = 8NUM_SUBCARRIERS = 64NUM_REALISATIONS = 1000POWER = 10NOISE = 1SEED = 42rng = np.random.default_rng(SEED)# Calculate Jain's fairness indexdef fairness_index(rates): denominator = NUM_USERS * np.sum(rates ** 2) if denominator == 0: return 0 return (np.sum(rates) ** 2) / denominator# Original channel-aware allocationdef channel_aware_allocation(rate_matrix): allocation = np.zeros(NUM_USERS) for subcarrier in range(NUM_SUBCARRIERS): user = np.argmax(rate_matrix[:, subcarrier]) allocation[user] += rate_matrix[user, subcarrier] return allocation

7.8 Experimental Results

After executing the Python program, the measured results should be recorded in the following table.

Table 7.1: Overall Performance Comparison

Performance Metric	**Original OFDMA**	**Redesigned OFDMA**
Average total throughput	Record result	Record result
Average Jain's fairness index	Record result	Record result
Average minimum user throughput	Record result	Record result
Throughput retention	100%	Record result

Fairness target achieved	Record result	Record result

Table 7.2: Individual User Throughput

User	Original Throughput	Redesigned Throughput
User 1	Record result	Record result
User 2	Record result	Record result
User 3	Record result	Record result
User 4	Record result	Record result
User 5	Record result	Record result
User 6	Record result	Record result
User 7	Record result	Record result
User 8	Record result	Record result

The values must be obtained by executing the simulation. They should not be estimated or manually invented.

7.9 Analysis of Results

The original channel-aware OFDMA method prioritises users with favourable channel conditions. Therefore, its main objective is to maximise total system throughput.

The redesigned method considers the current throughput of each user before allocating another subcarrier. This gives users with lower throughput a greater opportunity to receive additional resources.

The following points should be examined after obtaining the simulation results:

1. Whether the redesigned method increases Jain's fairness index.
2. Whether the minimum user throughput improves.
3. Whether the total throughput decreases compared with the original method.
4. Whether the redesigned method achieves the selected fairness target.
5. Whether the redesigned method retains at least 80% of the original total throughput.

If fairness improves while throughput remains above the selected target, the redesign can be considered successful for the tested conditions.

If the fairness target is achieved but total throughput falls below 80%, the allocation strategy may need further adjustment. Similarly, if throughput remains high but fairness does not improve sufficiently, a stronger fairness constraint may be required.

7.10 Validation of the Redesigned Solution

The redesigned strategy is validated by comparing it with the original method under identical simulation conditions.

The validation procedure is:

1. Generate 1,000 independent Rayleigh fading channel realisations.
2. Apply the original maximum-channel-gain allocation.
3. Apply the redesigned fairness-aware allocation to the same channel matrices.
4. Calculate the total throughput for both methods.
5. Calculate the average individual user throughput.
6. Calculate Jain's fairness index for every realisation.
7. Calculate the average minimum user throughput.
8. Compare the measured values against the performance targets.

The redesign is considered successful only if the measured results demonstrate the required improvement.

The experiment also produces a CSV file named ofdma_redesign_results.csv, which contains the average throughput of each user under both allocation methods.

7.11 Limitations of the Redesigned Strategy

Although the fairness-aware allocation improves the consideration given to users with lower throughput, it also has certain limitations.

1. It may reduce total system throughput because some subcarriers are allocated to users with weaker channel conditions.
2. It requires the scheduler to track the throughput of every user.
3. Frequent throughput calculations can increase scheduling complexity.
4. It does not guarantee equal throughput under every channel condition.
5. Users experiencing extremely poor channel quality may still receive limited data rates.

Therefore, the redesigned strategy provides a trade-off between system throughput and fairness rather than maximising both independently.

7.12 Further Improvements

The redesigned OFDMA scheduler can be improved further using proportional-fair scheduling.

Proportional-fair scheduling considers both the instantaneous channel quality and the average throughput received by each user.

A commonly used scheduling metric is:

\[PF_{i,k}=\frac{r_{i,k}}{R_i} \]

Where:

- \(PF_{i,k}\) = Proportional-fair scheduling metric.
- \(r_{i,k}\) = Instantaneous achievable rate of user \(i\) on subcarrier \(k\).
- \(R_i\) = Average throughput of user \(i\).

The scheduler can assign resources to users with a high instantaneous rate relative to their average throughput.

This approach provides a balance between exploiting good channel conditions and supporting users who have received fewer resources over time.

Other possible improvements include:

1. Minimum-rate guarantees for all users.
2. Priority-based scheduling for important users.
3. Adaptive resource allocation based on traffic demand.
4. Quality-of-Service-based scheduling.
5. Dynamic adjustment of fairness parameters.

7.13 Final Conclusion

This experiment examined the fairness limitation of channel-aware OFDMA allocation. The original strategy assigns subcarriers based mainly on channel quality, which can increase total throughput but may disadvantage users experiencing poor channel conditions.

A fairness-aware max-min allocation strategy was proposed and implemented using Python. The redesigned scheduler considers the throughput already received by each user before assigning additional subcarriers.

Both strategies were evaluated using the same eight-user, sixty-four-subcarrier system and 1,000 Rayleigh fading channel realisations.

The performance comparison considers total throughput, individual user throughput, minimum user throughput, and Jain's fairness index.

The final decision must be based on the measured experimental results. If the redesigned strategy improves fairness while retaining the required proportion of total throughput, it provides a better balance between efficient resource utilisation and fair service.

Thus, the experiment demonstrates the importance of diagnosing performance limitations, redesigning an existing solution, implementing the changes, and validating the improvements through measurable evidence.